

La Salle International School of Commerce and Digital Economy

Final Project

Graduate in Business Intelligence and Data Analytics

ENHANCING ENGAGEMENT AND USABILITY
IN AI-ASSISTED LEARNING:
A DATA-DRIVEN APPROACH USING POWER BI

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FINAL PROJECT DEFENCE

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Abstract

This project explores how student engagement, usability, and AI-assistant interaction develop within the MyWay language learning platform. By analyzing data extracted from over two million platform events with analysis in Python and visual integration in Power BI, the study uncovers meaningful patterns in student behavior. Key findings reveal usability drop-off points, behavioral clusters, and a promising impact of the AI assistant, the Kibot. The final deliverable is an interactive Power BI dashboard that provides stakeholders valuable insights to improve data-driven decision-making in digital education.

Resumen

Este proyecto analiza la participación, la usabilidad y el asistente virtual Kibot en la plataforma de aprendizaje de idiomas MyWay. A partir del análisis de más de dos millones de eventos, y mediante el uso de Python y visualizaciones en Power BI, el estudio identifica patrones significativos en el comportamiento del estudiante. Entre los principales hallazgos destacan puntos críticos de usabilidad, agrupaciones de perfiles de uso y un impacto prometedor del asistente virtual, el Kibot. El entregable final es un panel interactivo en Power BI que ofrece a los responsables información valiosa para mejorar la toma de decisiones basada en datos en el ámbito de la educación digital.

Resum

Aquest projecte analitza la participació, la usabilitat i la interacció amb l'assistent virtual Kibot dins de la plataforma d'aprenentatge d'idiomes MyWay. A partir de l'anàlisi de més de dos milions d'esdeveniments i la combinació de Python amb la integració visual a Power BI, l'estudi descobreix patrons rellevants en el comportament dels usuaris. Els principals resultats inclouen punts de caiguda en la usabilitat, agrupacions de perfils d'ús i un impacte prometedor de l'assistent virtual, el Kibot. El lliurable final és un dashboard interactiu a Power BI que ofereix als responsables informació útil per millorar la presa de decisions basada en dades en l'educació digital.

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Table of Contents

1.	Executive Summary	7
2.	Introduction.....	8
2.1	Background	9
2.2	Challenge Overview.....	10
2.2.1	Evaluating Usability and Design Effectiveness	10
2.2.2	Student Behavior and Engagement.....	11
2.2.3	Evaluating the Kibot's Impact	11
2.2.4	Methodology overview	11
2.2.5	Power BI	12
2.3	Data.....	13
2.3.1	Metrics Reports.....	13
2.3.2	Activity Reports.....	14
2.3.3	Data Anonymization and Format	14
2.4	Expected Results.....	14
3.	Theory	16
3.1	Digital Learning Platforms and Hybrid Learning.....	16
3.2	AI-Driven Tools in Education.....	17
3.3	Gamification in Education	19
3.4	Predictive Analytics in Education.....	19
3.5	Predictive Analytics and Business Intelligence (BI).....	20
3.6	Theoretical Frameworks	21
3.7	Gaps in the Literature and Conclusion.....	22
4.	Data.....	23
4.1	Metrics Dataset	23
4.2	Activities Dataset	24
4.3	Other Data Sources	25
4.4	Data Timeline and Privacy.....	26
5.	Methods.....	27
5.1	Initial Exploratory Data Analysis.....	27
5.1.1	Metrics report exploration.....	27
5.1.2	Activity reports exploration	29

5.2	Feature Engineering	31
5.3	Power BI dashboard design and implementation.....	32
5.3.1	Dashboard structure	32
5.3.2	Introduction (Report page 1).....	32
5.3.3	Usability Analysis Implementation (Report pages 2-5).....	33
5.3.4	Engagement Analysis Implementation (Report pages 6-8)	35
5.3.5	Kibot Analysis Implementation (Report pages 10-15)	36
5.4	Predictive Modeling integrated into Power BI	39
5.4.1	Model Design.....	39
5.4.2	Implementation in Power BI & utility	40
6.	Results.....	42
6.1	Usability Findings.....	42
6.2	Engagement Trends.....	45
6.3	Kibot Impact Analysis.....	46
6.4	Predictive Model Outcomes.....	47
7.	Discussion and Conclusions	49
7.1	Recap of results.....	49
7.2	Value of Power BI.....	49
7.3	Limitations	50
7.4	Recommendations and Future work	51
8.	Bibliography	52
9.	Appendix.....	57
9.1	Appendix A: Additional Figures	57
9.2	Appendix B: Additional Tables.....	62
9.3	Appendix D: Power BI pages overview.....	65
9.4	Appendix E: GitHub repository	70

List of Figures

Figure 1: MyWay platform world map with 4 missions	8
Figure 2: MyWay Mission 2 with 3 challenges, audio and playzone	8
Figure 3: Mission 1 overview with Kibot	9
Figure 4: Example of JSON "metrics report" structure	23
Figure 5: Example of JSON "activity report" structure	25
Figure 6: Data Timeline	26
Figure 7: Count of events by code	28
Figure 8: Event count through time	28
Figure 9: Count of moods	28
Figure 10: Distribution of moods.....	28
Figure 11: Overall score distribution	29
Figure 12: Duration distribution	29
Figure 13: Count of students per country	30
Figure 14: Linechart of number of activities throughout the year	30
Figure 15: Correlation between language skills.....	30
Figure 16: Overview of report page 1	32
Figure 17: Heatmap of Interactions example 1	33
Figure 18: Heatmap of Interactions example 2.....	33
Figure 20: Overview of report page 2.....	33
Figure 21: Sankey Diagram Example 1	34
Figure 22: Sankey Diagram Example 2	34
Figure 23: Report page 3 screen tooltip.....	34
Figure 24: Scatter plot with clusters report page 4	34
Figure 25: Overview of report page 4.....	35
Figure 26: Overview of report page 7.....	35
Figure 27: Report page 8 boxplot	36
Figure 28: Python script in Power BI visual.....	36
Figure 29: Report page 10 bar chart displaying frequency and trigger rates of AI Agent.....	36
Figure 30: Report page 11 line chart with weekly mood levels.....	37
Figure 31: Overview of report page 12.....	37
Figure 32: Report page 13 bar chart with AI messages	37
Figure 33: Report page 14 scatter plot of Kibot activations vs grade.....	38
Figure 34: Overview of report page 15.....	38
Figure 35: Correlation heatmap of features	39
Figure 36: Run Python button in Power Query	40
Figure 37: Overview of report page 9.....	40
Figure 38: Click interactions Mission 3 Challenge 1.....	42
Figure 39: Click interactions Mission 2 Challenge 1.....	42
Figure 40: Click interactions Mission 1	42
Figure 41: Screens with most "very high" density clicks	43
Figure 42: AudioPlayerScreen Click interactions.....	43

Figure 43: Unfiltered Sankey Diagram.....	43
Figure 44: Sankey Diagram filtered by destination	44
Figure 45: Sankey Diagram filtered by origin	44
Figure 46: Sankey Diagram filtered by User ID	44
Figure 47: Top 10 clicked screens for Over-clickers	45
Figure 48: Count of activites.....	45
Figure 49: Line chart of skill grades throughout time	45
Figure 50: Average score per country	46
Figure 51: Kibot-grade analysis for center ID 39	47
Figure 52: Conversion rate per estimated task.....	47
Figure 53: Feature importances	48
Figure 54: MyWay mission overview	57
Figure 55: MyWay world map overview	57
Figure 56: Power BI overview of opening page	58
Figure 57: Example of a page of a Power BI report	58
Figure 58: Data acquisition pipeline	59
Figure 59: User ID individual folders with interaction data	59
Figure 60: Timestamp compressed JSON folders inside a User ID folder	59
Figure 61: Language subcategories score distribution.....	60
Figure 62: Correlation heatmap of all variables	60
Figure 63: Model 1 results	61
Figure 64: Tooltip of activity count per user.....	61
Figure 65: Example 2 of Center ID kibot-grades analysis.....	61

List of Tables

Table 1: Most relevant variables of "Metrics reports". Complete table available in Appendix B: Additional Tables.....	27
Table 2: Most relevant variables of the activity reports. Complete table available on Appendix B: Additional Tables.....	29
Table 3: Features engineered throughout the study	31
Table 4: Tables created within Power BI	31
Table 5: Variables in metrics.csv.....	62
Table 6: Variables in activityreports.csv	63
Table 7: Summary of data sources	63
Table 8: Table of Models Comparison	64

WORD COUNT: 10981 words

1. Executive Summary

This project analyzes real student interactions with the MyWay learning platform, developed by Kids&Us in collaboration with Lighthouse DIG, with the goal of generating actionable insights on platform usability, student engagement, and the effectiveness of the AI assistant, the Kibot. The study is based on two primary datasets: one containing over 2 million user interaction logs (clicks, screen transitions, AI agent events), and another containing academic task completion and performance data. These raw files, extracted from Azure Blob Storage and internal performance reports, were parsed using Python, resulting in a series of datasets used in the analysis.

All findings are integrated into a 15-page interactive dashboard built in Power BI, designed for both technical and non-technical users. The dashboard is structured into four main areas. The Usability section uncovers navigation behaviors on the platform using techniques such as screen click heatmaps and Sankey diagrams. The Engagement section explores patterns in student performance, task duration, and usage trends by country and level. The Kibot section evaluates the assistant's message triggering and students' behavioral response to these. Finally, the Predictive Modeling section introduces a classifier capable of identifying whether a task is a student's first based on their interaction patterns, achieving over 91% accuracy with minimal features.

To do feature engineering, a combination of Python, Power Query, and DAX was used. Though the ambition was to keep the entire pipeline embedded in Power BI, technical limitations related to Python execution time and dataset size required some outputs (like the Kibot processing) to be done offline and imported. Despite these constraints, the report demonstrates the potential of combining business intelligence with lightweight machine learning to enhance learning platforms. This work offers a replicable blueprint for education-focused organizations who want to better understand and improve student experience through data.

2. Introduction

In recent years, digital learning platforms have become essential tools in education, transforming traditional learning paradigms. The global shift toward remote and hybrid learning models, accelerated by the COVID-19 pandemic, has now solidified the role of these platforms in modern education (Mulenga and Marbán, 2020). Digital learning environments offer flexibility, personalization, and accessibility, breaking down geographical and temporal barriers to education. (Faustmann et al., 2021) The integration of advanced technologies, such as artificial intelligence (AI), has enhanced the potential of digital learning environments to respond to individual student needs. (Khanduri and Teotia, 2023)

This research focuses on **MyWay**, an AI-assisted digital learning platform developed by Kids&Us, an English language learning company, in collaboration with Lighthouse DIG. MyWay is designed to complement traditional classroom learning by providing students with an immersive environment to reinforce their language skills. The platform features a gamified world map, where each continent represents a “mission” comprising reading, writing, speaking and listening activities (see Figure 1). Within each “mission”, students encounter “challenges” which are structured as puzzles or games, “audio activities”, and an optional “playzone” for additional practice after completing mandatory tasks (see Figure 2).

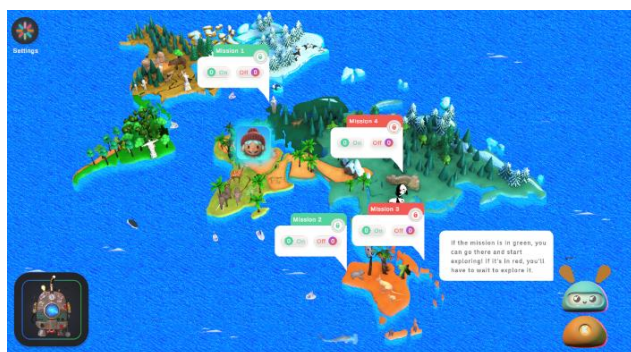


Figure 1: MyWay platform world map with 4 missions



Figure 2: MyWay Mission 2 with 3 challenges, audio and playzone

A core component of MyWay is the Kibot, an AI-powered assistant designed to improve the learning experience through personalized guidance and interaction. As seen in the image below (see Figure 3), it is located in the bottom right corner of the screen and displays messages to the student. The Kibot is programmed to analyze behavior using the platform and deliver targeted messages to encourage specific actions, such as completing tasks or revisiting challenge content. While the Kibot represents promising innovation in AI-driven education, its effectiveness in improving learning outcomes still needs to be empirically validated.



Figure 3: Mission 1 overview with Kibot

Other relevant images of the MyWay app are stored in Appendix A: Additional Figures. (Figure 53, Figure 54)

This study aims to explore how data-driven insights can be leveraged to track MyWay's overall performance. Specifically, it seeks to explore how student interaction data can be used to improve the platform's effectiveness.

2.1 Background

The rapid growth of digital learning environments has made the collection and analysis of student interaction data crucial for understanding patterns and optimizing learning experiences (Hase and Kuhl, 2024). However, tracking the efficiency in online settings remains a major challenge.

In response, BI tools have become vital in education, allowing institutions to transform raw, multi-source data into structured, actionable information. Organizations benefit from real-time visualizations and dashboards that allow educators to monitor patterns, detect friction points, and respond effectively (Chen, Chiang, and Storey, 2012; Williamson, 2015).

Power BI in particular stands out for its ability to integrate data from various sources and deliver interactive, intuitive visual reports. It enables both technical and non-technical stakeholders to gain insights into the matter. (Microsoft, 2024)

Understanding and supporting student behavior is crucial in educational settings, as emotional engagement can directly influence learning outcomes. However, the power of the information may remain underutilized if it is not presented in an accessible format. Interactive dashboards serve as a bridge between complex datasets and practical solutions, but their value depends on how well they are built with clear, interpretable visualizations. Research shows that the evolution of business intelligence increasingly demands tools that transform raw data into insights (Chen, Chiang, and Storey, 2012). In this context, Power BI plays a central role as the primary monitoring tool, enabling both real-time awareness and improvement actions.

Furthermore, in today's data-driven world, educational analysis has evolved to not only describe current trends but also anticipate future needs. BI tools can be extended with predictive models as a complementary layer to the visual narrative (Ryu, 2013).

Overall, tools like Power BI enable educational organizations to monitor and improve platform usage by combining visual clarity with data depth. (Long and Siemens, 2011). This project aims to develop such a tool to help digital learning platforms maximize engagement, usability, and learning effectiveness.

2.2 Challenge Overview

For Kids&Us and Lighthouse DIG, the challenge they face is not necessarily a problem in the traditional sense but a strategic need: to systematically evaluate learning behaviors. They seek to ensure that its design, engagement strategies, and AI-driven tools are effectively contributing to student learning. The company wants to evaluate three key aspects of the platform, all of which are explored through Power BI.

2.2.1 Evaluating Usability and Design Effectiveness

One of Kids&Us' primary concerns is whether the platform is user-friendly and accessible. Some of the metrics used to evaluate usability will be click interactions and screen navigation events. The effectiveness of digital learning is highly dependent on platform usability, as poor navigation and complex interfaces can frustrate the student, (Al-Mubireek, 2019) while interactive and gamified content increases retention and improves learning outcomes (Deterding et al., 2011).

To ensure the platform's effectiveness and accessibility, this project will assess whether students find the platform easy to use or whether there are barriers that affect their learning experience by analyzing the student's journey.

2.2.2 Student Behavior and Engagement

The second challenge is assessing if students are using the platform as intended to contribute to their learning. Engagement metrics such as completion rates, session durations, and interaction frequency are crucial indicators of student learning success. Research shows that student engagement in digital platforms is strongly influenced by the platform's ability to provide interactive and personalized formats. For example, platforms that incorporate gamification and immediate feedback mechanisms tend to foster higher levels of engagement and motivation (Deterding et al., 2011).

However, tracking engagement in online platforms remains a significant challenge. Studies indicate that there are struggles to stay motivated in digital environments (Hodges et al., 2020). By analyzing engagement metrics, this research aims to identify patterns of disengagement and provide recommendations for improving student retention and motivation.

2.2.3 Evaluating the Kibot's Impact

As an AI-driven virtual assistant, the Kibot is designed to support and enhance the learning experience, but its effectiveness in improving student learning has not yet been quantified.

The study will measure the Kibot interaction rates, as in how frequently the AI-companion actually assists the student, as well as a performance analysis to determine whether Kibot users show better results. Studies indicate that AI-driven tools can improve learning outcomes by adapting content to individual needs and providing targeted feedback (Khanduri et al., 2023). However, more research is needed to evaluate the long-term impact of these tools on student performance.

By analyzing these aspects, this project will determine whether the Kibot adds value to the learning process and how it can be further optimized.

2.2.4 Methodology overview

To answer these research questions, this study places Power BI at the core of its methodology. It seeks to go beyond a traditional BI dashboard, by building a dynamic, interactive environment integrating the necessary information for real-time decision-making.

The methodology for achieving this will include:

- Data Preparation (Python) where the raw data extracted from MyWay's platform is transformed from JSON format to tabular format. This is a critical step to ensure compatibility with Power BI. This step involved parsing nested structures and flattening event logs.
- Exploratory Data Analysis (EDA) to identify trends and patterns in student interaction data. This will involve analyzing click interactions, session durations and completion rates among others, to understand how students are using the platform. (Long and Siemens, 2011)
- Feature engineering, where new variables will be created and, if applies, selected.
- Power BI Dashboard Design and Implementation as the core of the project. The goal was the development of a comprehensive, user-friendly Power BI dashboards that offer stakeholders real-time insights.
- As a complementary analysis, predictive modeling to forecast engagement and usability issues. Machine learning algorithms can be used to predict which students are at risk of disengagement based on their interaction patterns (Arnold and Pistilli, 2012)

This study aims to go beyond individual interactions by analyzing aggregate data to identify long-term trends, predict future challenges, and provide actionable insights. The result will be a comprehensive report to provide a unified solution for monitoring the platform performance.

2.2.5 Power BI

Power BI is Microsoft's business intelligence and data visualization platform. It allows users to connect to a wide range of data sources, transform and model the data, and create interactive dashboards and reports.

Power BI was chosen as the primary analytical tool for the following reasons:

- Data integration capabilities: by allowing users to import and transform data from multiple formats.
- Powerful visuals, including heatmaps: Sankey diagrams, tooltips... crucial to reflect learning behavior.
- Accessibility and interactivity: enabling both technical and non-technical users to explore insights.
- Advanced Modeling: with support for DAX, custom measures and Python integration.

- Practical familiarity and growth: over time, Power BI became a central tool in my work with Lighthouse DIG, starting with basic dashboards in early projects evolving into a platform I fully leverage for real-time analysis and insights.

The main development work was done in Power BI Desktop, which allowed full control over data transformation, modeling and report design. The final dashboard can be shared through the Power BI Service, where it becomes accessible to a broader audience in a secure online environment. In addition, a mobile-optimized version of the dashboard was also developed. This allows key metrics and visualizations to be accessed easily from a mobile device, demonstrating the platform's flexibility across screen formats.

To illustrate key outputs, Power BI overview screenshots have been included in the Appendix A: Additional Figures (Figure 55, Figure 56).

Power BI provided the foundation for aggregating, exploring and communicating all aspects of the platform's performance, usability and AI effectiveness in a unified way.

2.3 Data

This study uses student interaction data from the MyWay digital platform. The data is collected through multiple sources, each serving a distinct purpose in analyzing both real-time interactions and long-term trends.

The most relevant data sources for this study are the following:

2.3.1 Metrics Reports

Metrics reports capture detailed logs of student interactions with the platform:

- Click interactions
- Screen navigation flows
- AI Agent events associated with the Kibot, such as message frequency and types.

These records provide granular insights into how students interact with the platform, including their navigation patterns and interactions.

The metrics reports are first saved in a temporary database called Redis, an in-memory storage service known for its ability to handle high-frequency data, required to capture user interactions in real-time. (Redis, 2023). After temporary storage in Redis, the data is transferred to Azure Blob Storage, a cloud-based storage solution designed for large-scale data storage and analysis

(Azure Blob Storage, 2023). Further information can be found in the Appendix (see Figure 57: Data acquisition pipeline)

2.3.2 Activity Reports

Activity reports summarize learning progress, include:

- Activities completed
- Time spent on the platform and the
- Scores achieved in different areas.

This data is crucial for understanding student progress and performance.

This data is stored in Cosmos DB, a NoSQL database designed for high-performance, scalable applications. It is particularly suited for handling large volumes of data with low latency (Azure Cosmos DB, 2023).

2.3.3 Data Anonymization and Format

Throughout this process, the data collected is anonymized to protect student privacy, in line with ethical data handling practices recommended in learning analytics research. (Slade and Prinsloo, 2013). The anonymized data is provided in either JSON or CSV format, depending on the final data export process. JSON format is commonly used for web-based data and allows for complex data structures, while CSV is a more straightforward tabular format compatible with most data analysis tools (Lantz, 2019). Once the data format is finalized, it will be imported into Power BI for further processing.

2.4 Expected Results

The main expected result of this study is to develop a **comprehensive, interactive business intelligence (BI) dashboard** using Power BI, to control and provide actionable insights for optimizing the MyWay platform.

More specifically, the dashboard is designed to:

- **Monitor student engagement** through behavioral metrics such as clicks, session length, and task flow. This will allow the company to intervene before disengagement occurs.
- **Evaluate platform usability** by visualizing screen transitions and identifying friction points.

- **Assess the impact of the Kibot** assistant and evaluate the long-term impact of it on learning outcomes.
- **Provide a centralized business intelligence solution**, enabling Kids&Us to make faster, data-driven decisions to enhance platform effectiveness.

In addition, the dashboard incorporates a **deeper layer developed in Python**. This component enhances the tool by offering analysis related to more complex data transformations.

3. Theory

This section explores the theoretical and practical aspects of digital learning platforms, AI-driven tools in education, gamification, predictive analytics, and its integration with business intelligence, focusing on their role in improving student engagement and learning outcomes.

These topics are supported by theoretical frameworks such as Behavior-Based Control Systems (BBCS) and Perceptual Control Theory (PCT), which provide a foundation for understanding how digital learning platforms can adapt to student behavior in real time.

3.1 Digital Learning Platforms and Hybrid Learning

Digital learning platforms have revolutionized education offering a range of tools and resources to support language learning. Remote learning allows students to access learning materials anytime and anywhere, enabling them to learn at their own pace (Al-Mubireek, 2019). This is particularly beneficial since it highlights the importance of the student and of self-paced learning (Mejía et al., 2024).

Furthermore, personalization is another key benefit, as platforms can adapt content to individual learning styles and paces, offering tailored experiences (Khanduri et al., 2023). According to Faustmann et al. (2019), user-centered design is a critical factor, as platforms must be intuitive, accessible, and easy to navigate to encourage adoption. Mejía et al. (2023) found that interactive features like gamification and real-time feedback increase students' intrinsic motivation, making learning more enjoyable and effective. This aligns with the design of MyWay, which prioritizes a seamless user experience to ensure being able to engage with the platform.

Scalability is also a significant benefit, as digital platforms can accommodate large numbers of students and make education more accessible (Long and Siemens, 2011).

Finally, data-driven decision-making, as platforms collect and analyze student interaction data to identify trends, predict outcomes, and implement targeted interventions (Long and Siemens, 2011).

However, one major issue digital platforms face is low engagement. A study performed by Alshubi, Tengku and Ashaari (2019) shows students often struggle to stay motivated in online environments, particularly when they feel isolated or disconnected from their peers. Many students still prefer face-to-face learning, which can make it difficult to maintain engagement in digital settings (Abozeid and Abdelaall, 2021)

Digital platforms can also be negatively influenced by poor usability. Complex or unintuitive platform designs can frustrate users and affect learning. A study by Al-Mubireek (2019) found that many language learning platforms suffer from poor navigation and lack of interactivity, which can reduce student engagement and satisfaction. According to Songkram et al. (2023), students' adoption of digital learning platforms is strongly influenced by perceived usefulness and perceived ease of use, as outlined in the Technology Acceptance Model (TAM). Students are more likely to engage with a platform if they believe it will help them achieve their learning goals and if they find it easy to use.

Technical issues also pose a significant challenge. Poor internet connectivity, platform glitches, and device compatibility issues can disrupt the learning experience (Hodges et al., 2020).

Additionally, ethical concerns arise from the collection and analysis of student interaction data (Slade and Prinsloo, 2013). Institutions must establish clear policies to ensure that student data is used ethically and responsibly. Furthermore, data usage in digital platforms at times remains underutilized, and teachers' attitudes towards data may sometimes be negative, as shown in the study carried out by Hase et al. (2022). According to their findings, a lack of trust, confidence in using digital tools or insufficient school support can lead teachers to rely on traditional methods rather than embracing the data collected by these platforms. This highlights the importance of a simple straightforward and easy to use tool that teachers can adopt without hesitation.

3.2 AI-Driven Tools in Education

AI-driven tools, such as virtual assistants, chatbots, and intelligent learning companions, have the potential to transform education, both how students learn and how educators manage the process, by providing personalized support, real-time feedback, and adaptive learning paths. However, their implementation comes both with opportunities and challenges.

The integration of AI in education has introduced innovative approaches to improving engagement, accessibility, and personalized learning experiences, and is increasingly being integrated into educational platforms. (Holmes, Bialik and Fadel, 2019) AI-driven companions and virtual assistants, such as the Kibot, which aim to provide real-time assistance and adaptive feedback, identify learning patterns to offer customized support, and encourage active participation through interactive elements. (Han et al., 2025; Khanduri and Teotia, 2023). While AI has shown promise in education, there is still limited empirical evidence on its true impact on student engagement and learning outcomes. (Woolf et al., 2013)

Liu, Li and Luo (2023) highlight that, in the context of English language teaching (ELT), AI can provide personalized feedback on grammar, vocabulary, and pronunciation, making language learning more effective. Similarly, Crompton et al. (2023) discuss the affordances of AI in ELT, such as adaptive learning systems that adjust content based on student performance. Another important role of AI is automating administrative tasks, such as grading and attendance tracking. This allows educators to focus more on teaching and less on administrative duties, improving overall efficiency (Woolf et al., 2023).

AI-driven tools also play a crucial role in providing emotional and motivational support to learners. Intelligent Learning Companion (ILC) systems, such as the Kibot, are a specific type of AI-driven tool designed to mimic human companionship by offering personalized feedback and encouragement (Han et al., 2023). These systems use algorithms to analyze student performance and adapt content to individual needs, enhancing both engagement and learning outcomes. For instance, the Kibot can detect when a student is struggling with a particular concept and provide targeted suggestions, such as revisiting the topic or taking a short break (Chou, Chan and Lin, 2003).

ILC systems can play several roles, including emotional support, recognizing and responding to learners' emotional states (frustration, boredom) (Chou, Chan and Lin, 2003), personalized feedback, offering tailored feedback based on the learner's performance and progress. This helps students understand their strengths and weaknesses and guides them toward improvement (Kim, 2007). Also, collaborative learning, facilitating interaction between learners and the system, as well as among learners themselves (Han et al., 2023). For instance, ILC systems can encourage group activities or peer discussions, fostering a sense of community (Han et al., 2023).

However, the implementation of AI in education presents several challenges. Ethical concerns are a significant issue, as its use raises questions about how student data is collected, stored, and used (Slade and Prinsloo, 2013), for example, how should platforms handle sensitive information about a student's emotional state. Institutions must establish clear policies to ensure that AI tools are ethically and responsibly employed. Platforms like MyWay must balance the benefits of predictive analytics and BI with the need to protect student data and ensure ethical use (Joseph et al., 2024). Another challenge is the lack of empirical evidence. While AI has shown promise in improving engagement and performance, more research is needed to evaluate its effectiveness over time (Khanduri et al., 2023). Additionally, current systems may lack the

sophistication to fully replicate human companionship or provide truly personalized support (Han et al., 2023).

3.3 Gamification in Education

Gamification, the integration of game-like elements in non-game contexts, has become a popular strategy for increasing student engagement in digital learning platforms. According to Yamba-Yugsi et al. (2023), by incorporating elements such as points, badges, leaderboards, and challenges, gamification transforms traditional learning into an interactive and enjoyable experience. This approach fosters a sense of achievement and competition, motivating students to actively participate and complete tasks.

One of the key benefits of gamification is its ability to make learning more engaging and enjoyable. Platforms that incorporate gamification elements create a sense of progression and accomplishment, encouraging students to complete their tasks (Deterding et al., 2011). For example, the world map in MyWay, where each continent represents a mission, provides students with a clear sense of direction and purpose, making learning more immersive.

Bennani, Maalel, and Ghezala (2021) further explore adaptive gamification, which tailors game elements to individual learners' preferences and behaviors. This approach addresses the limitations of one-size-fits-all gamification by personalizing the learning experience. For example, adaptive systems can adjust difficulty levels or reward structures based on student performance, ensuring that all learners remain engaged and challenged.

But, as it's been emphasized, implementing adaptive gamification requires sophisticated algorithms and data analysis, which can be resource intensive. Its design requires careful planning to ensure alignment with educational goals and student needs (Smith and Johnson, 2022). Also, Erbaşı, Tural and Çoşkuner (2023) state that gamification must be carefully outlined to avoid superficial engagement, where students focus on earning rewards rather than understanding the material.

3.4 Predictive Analytics in Education

Predictive analytics has emerged as a powerful tool in education, enabling institutions to forecast student performance and provide actionable insights that help educators intervene and tailor learning experiences to individual needs.

One of its primary applications is identifying at-risk groups. By analyzing engagement metrics such as login frequency, assignment completion rates, and quiz scores, predictive models can

flag students who are likely to drop out or underperform (Arnold and Pistilli, 2012). For example, platforms like MyWay use predictive analytics to monitor student engagement and provide real-time feedback, helping students stay on track (Khanduri et al., 2023).

Another key role of predictive analytics is optimizing learning paths. By analyzing historical data, predictive models can recommend personalized learning paths based on student performance and preferences (Chen et al., 2012). For instance, if a student struggles with a particular concept, the system can suggest additional resources or alternative learning activities to address the gap.

Several techniques are used in predictive analytics to analyze educational data and generate insights. These include machine learning algorithms, such as decision trees, random forests, and neural networks, which are commonly used to predict student outcomes based on historical data (Munir, Vogel and Jacobsson, 2022). Data mining techniques are also used to extract patterns and trends from large datasets, such as grouping students based on their learning behaviors (Munir, Vogel and Jacobsson, 2022). Additionally, deep learning frameworks, such as TensorFlow and PyTorch, are increasingly being used to analyze complex educational data, such as natural language processing (NLP) for analyzing student essays or discussion forum posts (Munir, Vogel and Jacobsson, 2022). These frameworks can provide deeper insights into student behavior and engagement.

Despite its potential, predictive analytics in education present several challenges, one of which is data quality. Inaccurate or incomplete data can lead to wrong predictions, undermining the effectiveness of predictive models (Long and Siemens, 2011). Furthermore, there might be ethical concerns. The use of student data raises questions about privacy, consent, and algorithmic bias (Slade and Prinsloo, 2013). Institutions must establish clear policies to ensure that data is used ethically and responsibly. Finally, implementing predictive analytics requires technical expertise and infrastructure, which may be lacking in some institutions (Munir, Vogel and Jacobsson, 2022).

3.5 Predictive Analytics and Business Intelligence (BI)

Predictive analytics and business intelligence (BI) are closely related fields that leverage data to drive decision-making and optimize outcomes. While predictive analytics focuses on forecasting future events, BI focuses on analyzing historical data to provide insights into past performance. Together, these tools can provide a comprehensive view of organizational

performance and enable data-driven decision-making at the institutional level (Chen et al., 2012).

Predictive analytics plays a crucial role in BI by enabling organizations to forecast future trends and make proactive decisions. For example, in the banking sector, predictive analytics is used to forecast customer behavior, detect fraud, and optimize financial decision-making (Abhulimen et al., 2024). Similarly, in education, predictive analytics can be integrated with BI tools to provide information about student performance and institutional effectiveness.

By providing insights into past performance and future trends, predictive analytics and BI enable organizations to make data-driven decisions (Abhulimen et al., 2024). In addition, BI tools can automate data analysis and reporting, reducing the time and effort required to generate insights (Abhulimen et al., 2024). And as a result, optimized outcomes, predictive analytics can identify opportunities for improvement and recommend actions to optimize outcomes (Munir, Vogel and Jacobsson, 2022).

On the downside, implementing predictive analytics and BI also requires technical expertise and infrastructure, which may be lacking in some organizations (Lee, Cheang and Moslehpour, 2022), and as with any activity relating sensitive data, the use of it raises questions about privacy, consent, and algorithmic bias (Slade and Prinsloo, 2013).

3.6 Theoretical Frameworks

This section introduces the theoretical frameworks that underpin the research on digital learning platforms and AI-driven tools: Behavior-Based Control Systems (BBCS) and Perceptual Control Theory (PCT).

BBCS are adaptive systems designed to monitor and adjust user behavior to maintain optimal engagement levels. In digital learning platforms, BBCS can analyze student interactions and provide real-time feedback to encourage desired behaviors (Arkin, 1998). For example, if a student is disengaged, the system can suggest alternative activities or adjust the difficulty level of tasks to re-engage the student.

PCT explains behavior as a process of controlling perception. In the context of digital learning, PCT can be used to model how students adjust their actions to achieve desired learning outcomes (Powers, 2005). For example, if a student is struggling with a particular concept, they may adjust their study habits or seek additional resources to improve their understanding.

By integrating BBCS and PCT, this research aims to develop a unified solution that leverages predictive analytics and BI to monitor and enhance platform performance. These frameworks provide a foundation for understanding how digital learning platforms can adapt to student behavior in real time, enabling proactive interventions and long-term optimization.

3.7 Gaps in the Literature and Conclusion

Despite advancements in digital learning platforms, AI-driven tools, gamification, and predictive analytics, significant gaps remain. First, there is limited empirical evidence on the long-term effectiveness of AI-driven virtual assistants like the Kibot. While these tools show promise in providing real-time feedback, their impact on sustained learning outcomes is not evident (Khanduri et al., 2023). Second, while gamification enhances engagement, its long-term effects on motivation and deep learning are poorly understood (Erbaş, Tural, & Çoşkun, 2023). Third, ethical concerns of predictive analytics, such as privacy, consent, and algorithmic bias, lack of frameworks for responsible implementation (Slade and Prinsloo, 2013). Fourth, the integration of predictive analytics with business intelligence (BI) in education is underexplored, particularly in creating unified tools for actionable insights (Chen et al., 2012). Finally, there is a lack of research on teacher and institutional adoption of data-driven tools (Hase et al., 2022).

This study addresses these gaps by empirically validating the Kibot's impact, exploring long-term gamification effects and integrating predictive analytics with BI dashboards. The theoretical frameworks of Behavior-Based Control Systems (BBCS) and Perceptual Control Theory (PCT) provide a foundation for understanding how platforms like MyWay adapt to student behavior in real time. BBCS enables real-time adjustments, while PCT explains how students perceive and respond to these changes.

4. Data

This section presents the data used in this study in greater depth, detailing its acquisition, and structuring processes. The dataset reflects real student engagement in the MyWay platform. As previously introduced in Section 2.3, the data used is split into two main sources:

- 1- Metrics reports, which includes logs of platform use by students.
- 2- Activity reports, which contain task-level academic performance data.

Each dataset followed a different acquisition pipeline and required specific extraction techniques. Together, these datasets offer a view of both student behavior and student performance. From these two foundational sources, other purpose-specific datasets were created using a combination of Python scripts and Power BI transformations.

4.1 Metrics Dataset

The main dataset used originates from Azure Blob Storage, with anonymized logs of approximately 1,600 students. The data is initially downloaded in folders per user ID, where within each folder, multiple JSON files represent events collected over time. (see structure in Appendix A: Additional Figures, Figure 57) These files contain fields such as event type, timestamp, screen name, user ID, stress and engagement indicators, and Kibot interaction data.

```
{
  "user_events": [
    {
      "id": "1609459200-0",
      "event": {
        "timestamp": "2024-08-12T12:00:00Z",
        "sessionId": 123456,
        "userId": "user123",
        "device": "android",
        "environment": "pro",
        "type": "INFO",
        "code": "CLICK",
        "message_x": 100,
        "message_y": 200,
        "message_type": "click"
      }
    }
  ],
  "user_state": {
    "mood": "neutral",
    "no_progress_streak": 0,
    "last_activity_difficulty": 0.5,
    "last_activity_skill": "writing"
  },
  "mood_levels": {
    "stress_level": 0.2,
    "bored_level": 0.1,
    "engaged_level": 0.7,
    "frustrated_level": 0.0,
    "neutral_level": 0.0
  }
}
```

Figure 4: Example of JSON "metrics report" structure

An example of this JSON structure can be seen in Figure 4, where a single event is recorded with nested data under "user_events", "user_state" and "mood_levels".

Within user_events there are 3 possible types of events registered under the variable CODE: click events (x y coordinates of the interaction), screen changes events (details of from which screen to which), and AI Agent events (details of the Kibot's triggering, the AI-companion).

User_state and mood_levels give further context on emotional and cognitive user states at the time of the event.

To analyze this data, a custom Python notebook *processingmoods.ipynb* was developed using standard libraries (pandas, json, os) in order to:

- Extract data from each folder and file
- Parse and flatten the JSON structures
- Convert the nested structure into a single, tabular format (Pandas DataFrame).
- Export the clean dataset into a CSV file.

The resulting data is stored in *metrics.csv* and contains more than 2 million rows, with a final size of approximately 1GB. This file was successfully imported into Power BI Desktop.

From *metrics.csv*, other datasets were developed within python with different purposes within the analysis.

- *Moods.csv*, which contains only the data from *user_state* and *moods_levels* from Figure 4, excluding *user_events*. This is useful to extract the state of the individual at each specific timestamp.
- *Kibot_events.csv*, which contains filtered data from metrics where the type of event has been an AI Agent event. (CODE = AI_Agent)
- *Screens_summary.csv*
- *Repeated_user_summary.csv*

4.2 Activities Dataset

Unlike the metrics dataset, the activity reports dataset required a more manual acquisition process. Due to data governance restrictions by Kids&Us, the initial access was limited to a sample of 10 anonymized students. The files were exported and shared by a colleague, although the precise technical steps for the export remain undocumented. At a further stage of the project, the data of the same 1600 user IDs became available.

The data was stored in a folder containing multiple JSON files, each file representing one student's performance report. These reports include metrics such as overall task completion, vocabulary, grammar, and reading scores and activity durations.

Figure 5 shows a representative sample of the activity report JSON format.

```
user:
  id: string
  groupId: integer
  centerId: integer
  country: string

activity:
  id: string
  type: string
  location:
    missionId: string
    challengeId: string
    journeyId: string
    stepId: string

evaluation:
  overallScore: integer
  level: integer
  duration: integer
  skills:
    listening: integer
```

Figure 5: Example of JSON
"activity report" structure

A similar process was used with a custom Python notebook, *processingactivities.ipynb*, where data was parsed and converted to a clean standardized format compatible with the metrics dataset.

This resulted in *activityreports.csv*, which, at its turn, allowed us to extract:

- *Activities_with_moods.csv*, where each activity was matched with the *moods.csv*, retrieving each user's state around the moment of the completion of the activity.
- *Kibot_with_grades.csv*, as a merge between *activityreports.csv* and *kibot_events.csv* where more complex calculations allowed us to retrieve the average score per number of Kibot activations.

4.3 Other Data Sources

In addition to the core datasets and those created from them, two smaller resources were provided by Lighthouse DIG and integrated into the dashboard:

- *imgs.txt*, containing links to images of each platform screen. These were matched with screen names in the metrics tables to allow for screen data to interact with the corresponding image.
- *screen_transitions_click_frequency.csv*, detailing average time spent on each page. This data was used to generate comparative bar graphs and evaluate usability based on time engagement.

Table 7: Summary of data sources synthesizes the data used in this project. All notebooks can be found in the Appendix E: GitHub repository.

4.4 Data Timeline and Privacy

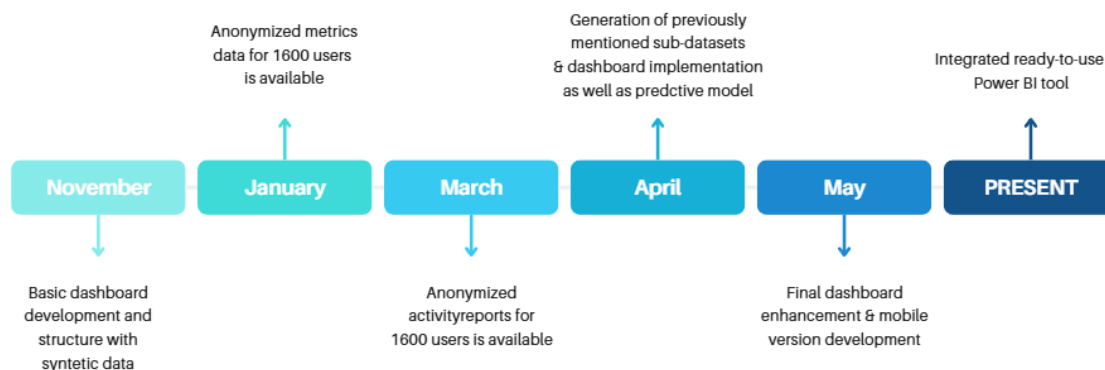


Figure 6: Data Timeline

As seen in Figure 6 above, during the early stages of the project, from November to January, no real data was available. To begin with development and design of the dashboards and models, synthetic data was generated using predefined structures. This step allowed early experimentation with the framework before integrating real usage data.

Data privacy and ethical usage were prioritized throughout the project. All datasets were fully anonymized in accordance with internal data protection policies and aligned with ethical standards in learning analytics research (Slade & Prinsloo, 2013). Identifiable fields such as usernames, emails, or school codes were either removed or encrypted before analysis. This step was influential in determining when the data was ready to be used.

By January, the real metrics dataset became available, which included anonymized interaction data for approximately 1600 students, covering the period from September to January – the months during which the MyWay platform had already been active and in use by learners.

As for the activity reports, the data was acquired manually in the month of March.

5. Methods

The main contribution of this project is the creation of a Power BI report that integrates and contextualizes real usage data, allowing us to identify patterns and evaluate efficiency. This section outlines the full methodological approach used to achieve this.

The study followed a logical pipeline from data acquisition (explained in Section 0), on to exploratory data analysis (EDA), feature engineering, dashboard development, and concluding with the integration of a predictive model using embedded Python scripts.

A strong emphasis was placed on creating a seamless experience for both technical and non-technical users, ensuring that the insights were both interpretable for stakeholders at Kids&Us and Lighthouse.

5.1 Initial Exploratory Data Analysis

The first step after data extraction and transformation (as discussed in Section 0), was to explore the structure and contents of the dataset to understand completeness and potential of the data. EDA was primarily conducted in Python, mainly using Pandas, Matplotlib and Seaborn.

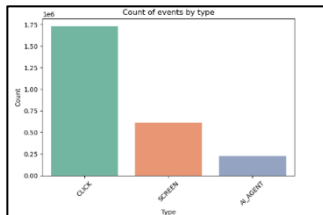
5.1.1 Metrics report exploration

To understand the context of the data, the following table shows the most relevant variables for our study found in *metrics.csv*:

Table 1: Most relevant variables of "Metrics reports". Complete table available in Appendix B: Additional Tables

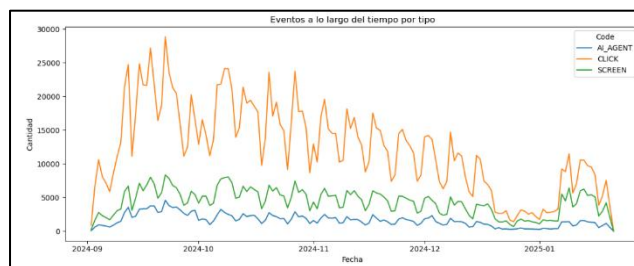
VARIABLE	DESCRIPTION
Timestamp	Event time and date
UserID	Anonymized unique student identifier
Code	Type of event: Click, Screen or AI Agent
Mood	Current estimated mood of student
Stress_Level	Stress level of student (0-1)
Bored_Level	Bored level of student (0-1)
Message_X	X coordinate of Click events
Message_Y	Y coordinate of Click events
Message_From	Initial screen of Screen events
Message_To	End screen of Screen events
Last_Screen	Screen where the interaction happens
Message_Anchor	Type of Kibot interaction
Message_Triggered	Whether the Kibot's message was triggered
Previous_Task	Task the student has just performed
Current_Task	Task estimated to do next

The metrics dataset contains over 2 million event rows, each representing a user interaction such as clicks, screen visits, or AI interactions, along with mood and engagement metrics.



During this phase of the research, I visualized the distribution of event types. As seen in Figure 6, clicks on the screen are, by far, the most recorded type of event. On the other hand, AI Agent events are the least frequent (Figure 7).

Figure 7: Count of events by code



Another insight of early analysis was related to engagement over time, which peaked during the initial months of the launch, and was then followed by a steady decline. (Figure 8)

Figure 8: Event count through time

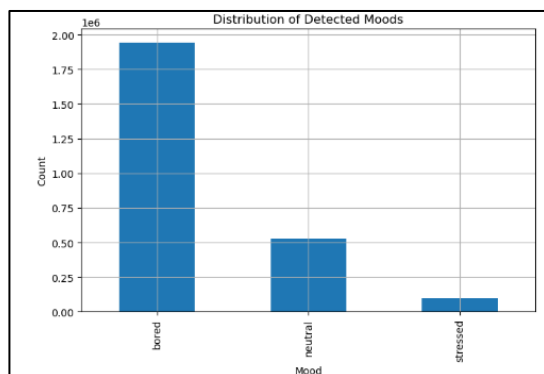


Figure 9: Count of moods

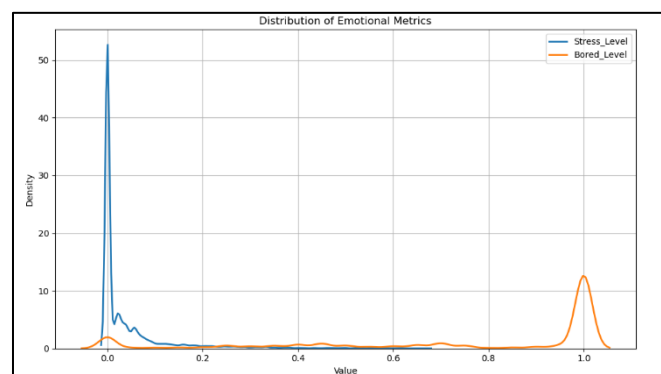


Figure 10: Distribution of moods

Importantly, analysis of the mood variable revealed that “bored” was detected in over 75% of observations (Figure 9), while the distribution of emotional levels (Figure 10) was far from normal. This led to a strategic shift: moods would still be analyzed but not treated as the main variable of interest.

These initial insights helped determine which metrics were worth engineering and which behaviors would be valuable to monitor or predict. All related scripts are included in the *usability.ipynb* and *kibot.ipynb* notebooks in Appendix E: GitHub repository.

5.1.2 Activity reports exploration

The following table shows the most important variables found in the *activityreports.csv* dataset:

Table 2: Most relevant variables of the activity reports. Complete table available on Appendix B: Additional Tables

VARIABLE	DESCRIPTION
User.id	Anonymized unique student identifier
User.courseID	Course the student is enrolled in
User.Country	Country associated with the student
Activity.id	Unique identifier for each activity
Activity.type	Type of activity (mission or playzone)
Activity.isFirst	Boolean (T/F) indicates if it's the first attempt
Evaluation.level	Level of difficulty of the activity
Evaluation.overallScore	Final score of the activity
Evaluation.duration	Duration of the activity in seconds
createdAt	Timestamp when the activity was submitted

The activity reports dataset contains 250K rows, each representing a task that has been completed by a student., either mission or playzone.

In this section, some distributions have been visualized:

- Overall score
- Evaluation duration
- Individual skill evaluations (grammar, vocabulary...)

The results indicated that most score-related variables were right skewed, suggesting that most students performed well (Figure 11). However, duration's distribution is left-skewed, hinting at either lack of difficulty or issues in evaluation design, for example (Figure 12).

Among the six evaluated skills, grammar and vocabulary were the most frequently practiced (based on row count), while speaking and writing showed flatter distributions, indicating higher difficulty or student avoidance (Appendix A: Additional Figures, Figure 60).

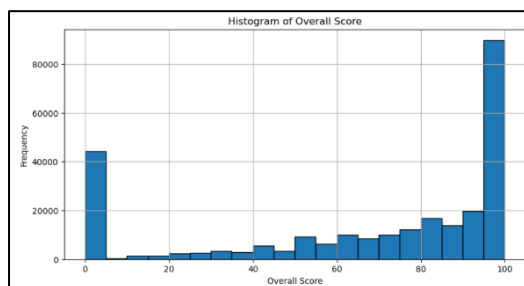


Figure 11: Overall score distribution

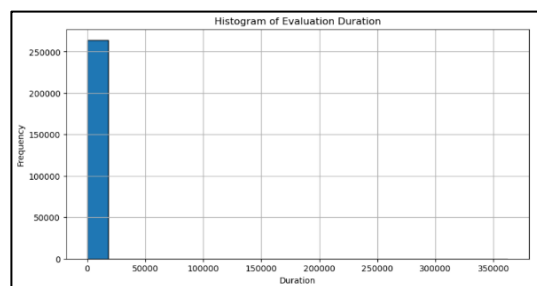


Figure 12: Duration distribution

To further contextualize, I plotted the number of unique students per country. Spain had the highest representation, an important consideration for the analysis. This dominance makes sense given the origin of Kids&Us, and the spread suggests potential for scalability.

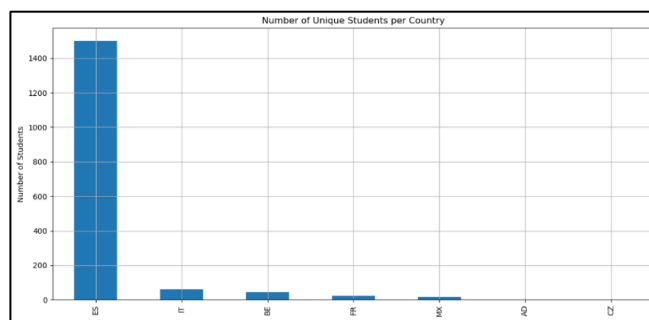


Figure 13: Count of students per country

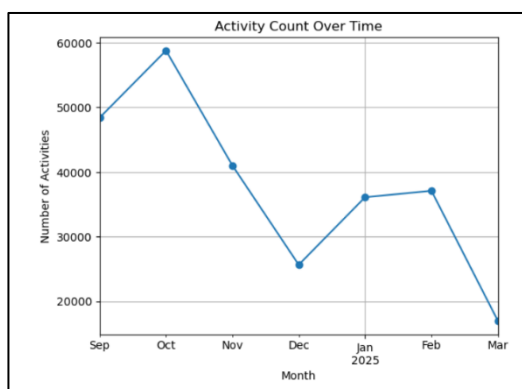


Figure 14: Linechart of number of activities throughout the year

A time series of activity completion confirmed that platform usage followed a launch spike followed by gradual decline.

The peak in early months aligns with the start of the academic year, and the dip in December is likely related to school holidays.

Finally, a correlation heatmap (Figure 15) showed strong independence between skill scores, especially for vocabulary, which was positively correlated with reading, listening and grammar.

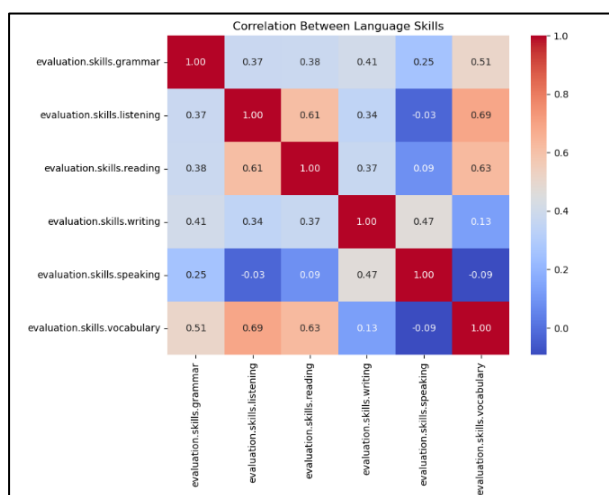


Figure 15: Correlation between language skills

5.2 Feature Engineering

After the EDA, the next step involved designing a series of features that could enhance the quality of insights and the interpretability of the data. This process, known as feature engineering, was critical to simplifying large volumes of data.

These variables were created using a combination of tools: Python (for complex logic), within Power BI's Power Query Editor or with calculations using DAX (Data Analysis Expressions). They served both descriptive and predictive purposes.

The following tables provide an overview of the main features and tables developed across the study:

Table 3: Features engineered throughout the study

FEATURE NAME	DESCRIPTION	ORIGIN
Last Screen	Last Screen where the interaction happened	Python (processingmoods.ipynb)
Time to next	Time, in seconds, until the next event by user	Python (usability.ipynb)
Confused_clicks	Number of clicks in a screen without further navigation during > 60s	Python (usability.ipynb)
Conversion success	Whether the student completed the action suggested by the Kibot within 10 minutes	Python (kibot.ipynb)
ComputedDensity	Normalized value based on click frequency	PowerQuery
Year-Week	Week of the year from timestamp	DAX
TopMoodCategory	Most frequent reported mood during a session	DAX
PointDensityColor	Color code assigned to each click by density	DAX
User clusters	Grouping based on click interactions and duration	Power BI

Table 4: Tables created within Power BI

TABLE NAME	DESCRIPTION	ORIGIN
Legend Table	Reference table linking colors to density labels	PowerQuery
Select_Metric	Reference table to choose between two metrics	PowerQuery

5.3 Power BI dashboard design and implementation

The analytical core of the project is the Power BI dashboard, which gathers insights generated from all datasets, and all features engineered, unified into an interactive, and stakeholder-friendly interface.

5.3.1 Dashboard structure

The final Power BI report is structured into 15 pages, which are grouped into 5 categories:

- **Introduction:** Overview with KPIs, monthly usage trends and event breakdown. It provides a high-level view of dataset statistics.
- **Usability analysis I-IV:** Focused on navigation patterns, screen visit behavior, click densities, and time spent on screens.
- **Engagement I-IV:** Examines student performance and displays metrics related to task completion, duration and grades. This section includes the predictive model.
- **Kibot I-VI:** Dedicated to the Kibot assistant's role, exploring its message frequency rate, conversion success and impact on academic performance.
- **Screen Tooltip:** Provides a tooltip with screen images linked to the text.

5.3.2 Introduction (Report page 1)

The first page gives an overview of the data. It displays metrics such as the number of students, the number of activities and the number of interactions per type, along with a timeline of platform usage. A navigation pane located in the bottom right corner allows the user to quickly navigate to the sections on usability, engagement or Kibot performance (Figure 16).

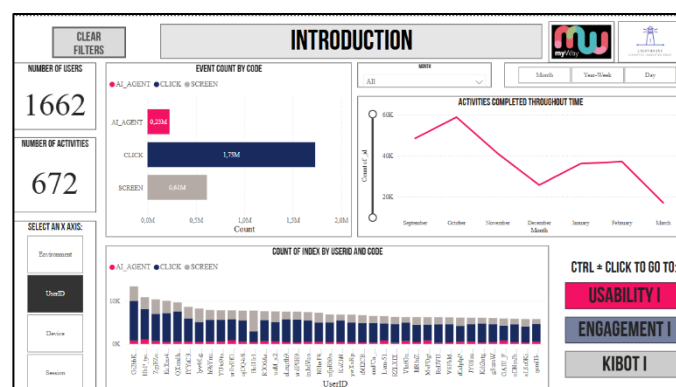


Figure 16: Overview of report page 1

Additionally, a custom JSON theme file was imported into Power BI to align the dashboard with the corporate branding of Lighthouse DIG. This included the use of the corporate font, as well as the colors across all pages to ensure visual coherence.

5.3.3 Usability Analysis Implementation (Report pages 2-5)

This section aims to visualize how students interact with the interface, encompassing pages 2-5 of the report.

One of the most visually engaging components is the click heatmap overlaid on actual platform screen images (Figure 17, Figure 18). This feature allows users to select a specific screen and instantly see where students clicked the most. Click coordinates are plotted directly over the background image, which are linked to screen names using the *imgs.txt* dataset in combination with *metrics.csv*. The result helped reveal which buttons attracted attention and which areas were rarely explored.

Figure 19 shows an overview of what the whole page looks like. The user can filter by screen, and, optionally, by a specific user ID. Above the heatmap, key interaction metrics are displayed, along with an interactive legend that filters by click density. This constitutes the “Usability I” page of the report.

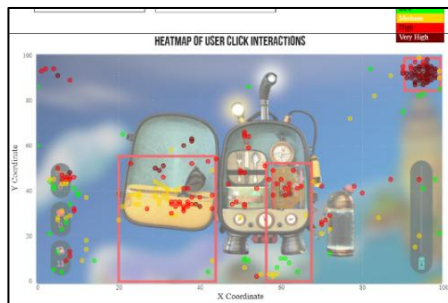


Figure 17: Heatmap of Interactions example 1

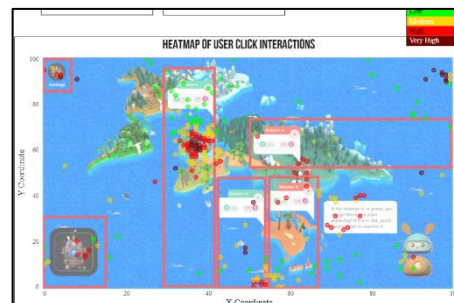


Figure 18: Heatmap of Interactions example 2

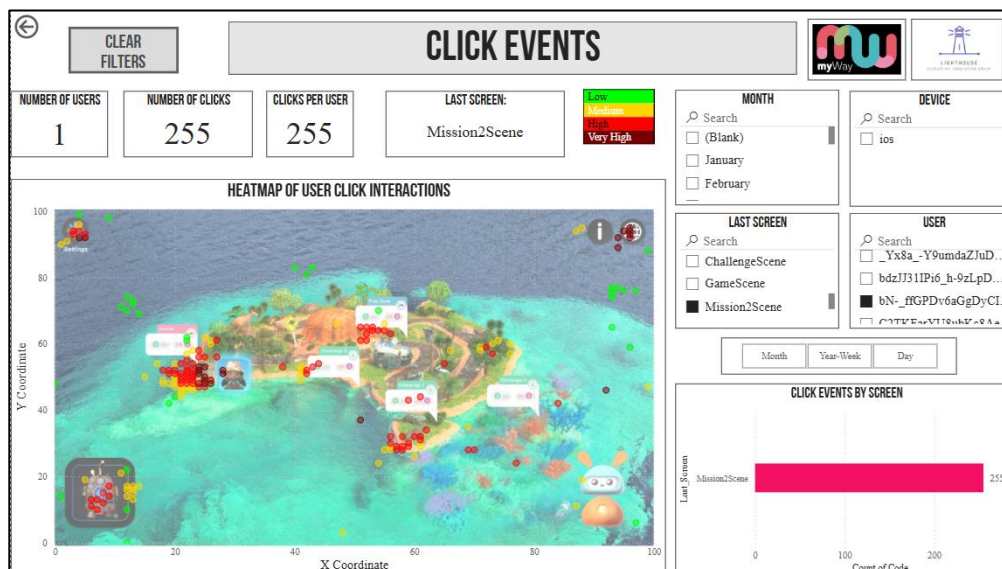


Figure 19: Overview of report page 2

To analyze user navigation through the platform, a screen transition diagram was created using a Sankey visual. This shows viewers the most common paths taken by users, such as movements from the World Map to a Mission or from a Challenge to the Playzone, and it highlights points where students commonly exited or looped back to previous screens.

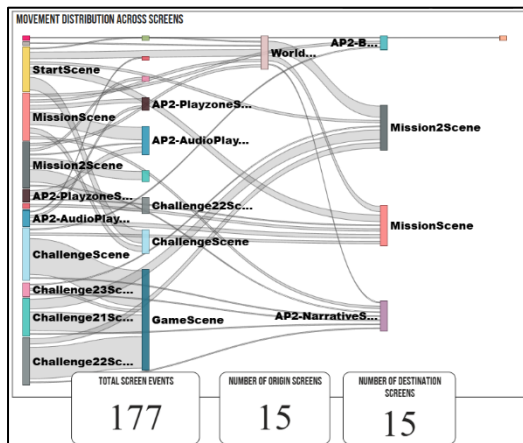


Figure 20: Sankey Diagram Example 1

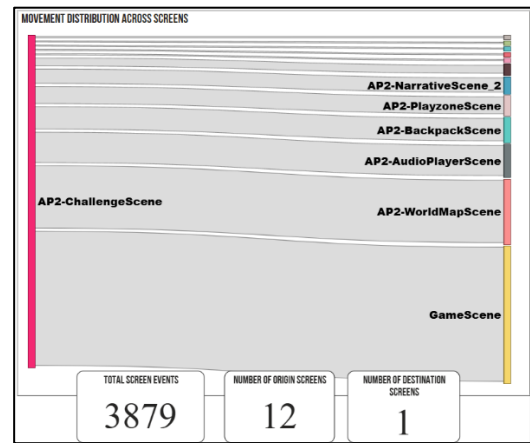


Figure 21: Sankey Diagram Example 2

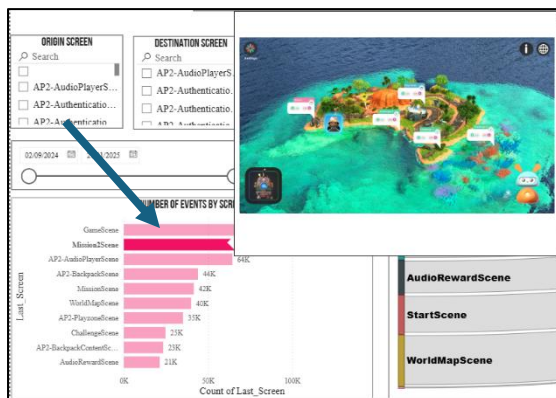


Figure 22: Report page 3 screen tooltip

On the “Usability II” page, a bar chart was developed showing the number of events per screen (Figure 22).

This includes a tooltip from page 16 of the report, which displays the image of the screen when hovered over, helping users contextualize the screen names.

On Usability III, the relationship between click frequency and time spent was analyzed using the *screen_transitions_click_frequency.csv* file. With clustering functionalities embedded in Power BI, users were grouped based on behavior (Figure 23).

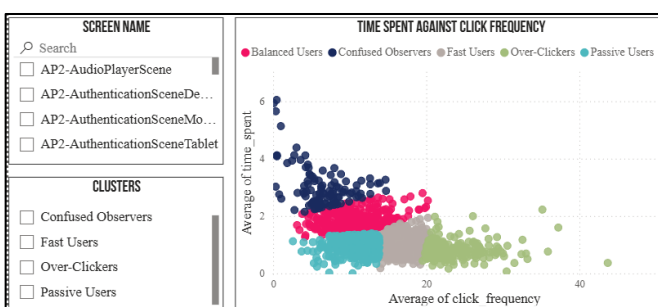


Figure 23: Scatter plot with clusters report page 4

Clusters were renamed to reflect observed usage patterns, for instance, confused observers tend to spend more time with fewer clicks, while over-clickers have short sessions with a high click intensity.

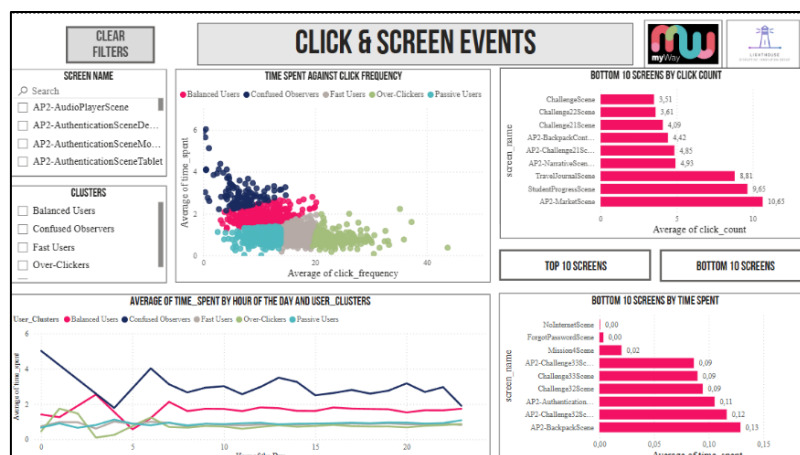


Figure 24: Overview of report page 4

This page also allows toggling between the top 10 and bottom 10 most visited screens.

This has been developed using the “Bookmarks” and the “Selection” features within Power BI.

5.3.4 Engagement Analysis Implementation (Report pages 6-8)

The engagement section is comprised of four pages, one of which is dedicated to the predictive model that will be explained separately in page 39.

The “Engagement I” serves as a general summary with simple visualizations such as bar charts of most performed activities or pie charts with proportions of activity types done. These results will be discussed in more depth in Section 6.

“Engagement II” page goes deeper into demographic analysis and grades. As mentioned in the EDA section, Spain has the highest number of users. This page compares performance across countries and examines variations in scores across evaluation categories such as reading, writing, grammar and vocabulary.

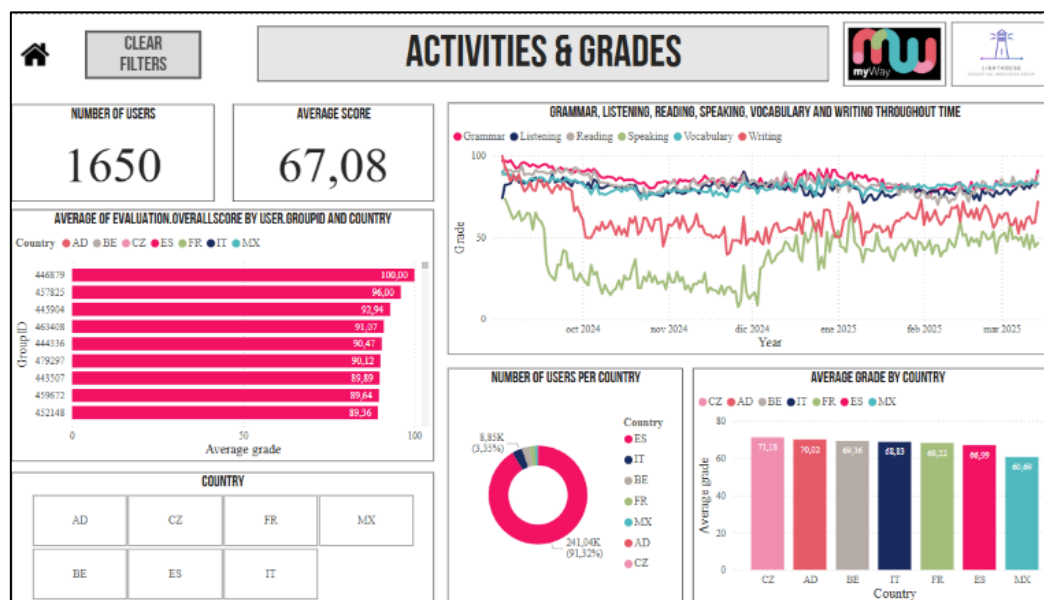


Figure 25: Overview of report page 7

The “Engagement III” page integrates Python-based visualizations using Power BI’s embedded scripting editor. Although visually it is less aesthetic, despite its simplicity, this page enables advanced analytical conclusions into user engagement patterns.

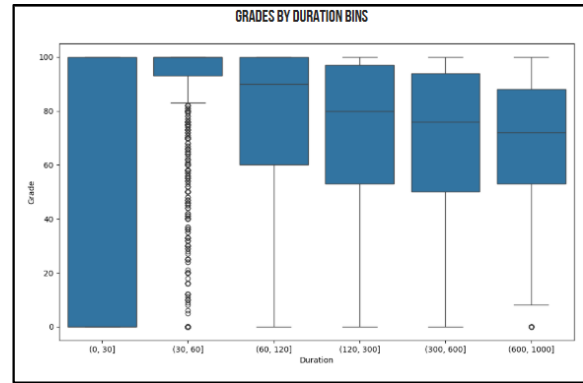


Figure 26: Report page 8 boxplot

Figure 27 shows how the Python editor interface looks within the Power BI environment.

```
Python script editor
△ Duplicate rows will be removed from the data.
6 # Paste or type your script code here:
7
8 import pandas as pd
9 import matplotlib.pyplot as plt
10 import seaborn as sns
11 dataset['duration_bins'] = pd.cut(dataset['evaluation.duration'], bins=[
12     plt.figure(figsize=(10, 6))
13     sns.boxplot(data=dataset, x='duration_bins', y='evaluation.overallScore'
```

Figure 27: Python script in Power BI visual

5.3.5 Kibot Analysis Implementation (Report pages 10-15)

This section is dedicated to exploring the implementation of the six pages of the Kibot AI assistant.

Page “Kibot I” presents an overview of the AI Agent message activity through a stacked bar graph, showing how often Kibot messages were triggered. This helped assess whether the Kibot's messages were effective or potentially intrusive.

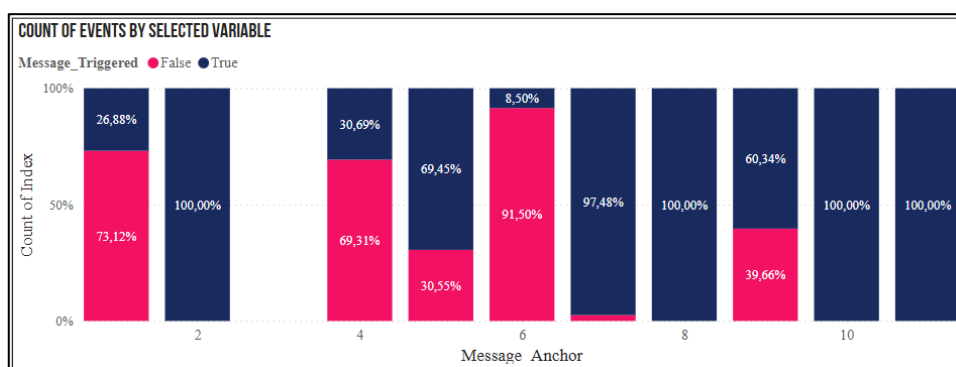


Figure 28: Report page 10 bar chart displaying frequency and trigger rates of AI Agent

“Kibot II” and “Kibot III”, provide a deeper analysis on mood and task estimator, which are closely related to how the Kibot performs afterwards. Although mood is not a primary focus in this study (due to data imbalance), it remains relevant for understanding the emotional state

AVERAGE MOOD VALUES THROUGH TIME

Legend: Bored_Level (Red), Stress_Level (Dark Blue), Neutral_Level (Light Blue), Frustrated_Level (Green), Engaged_Level (Teal)

Y-axis: 0.0, 0.5, 1.0

X-axis: Year-Week (2024-36 to 2025-4)

Buttons: Month, Year-Week (selected), Day

TASK ESTIMATOR

SELECT A PREVIOUS TASK:

- ☒ DoNothing
- ☐ GoToAudio
- ☐ GoToBackpack
- ☐ GoToMarket
- ☐ GoToPlayzone
- ☐ GoToStep
- ☐ GoToTravelJournal

TASKS THROUGHOUT TIME

Legend: ● DoNothing ● GoToAudio ● GoToBackpack ● GoToMarket ● GoToPlayzone ● GoToStep ● GoToTravelJournal

Time Range: Month Year Week Day

bored
TopMoodCategory

Current_Task	%GT Count of Current_Task
DoNothing	3.14%
GoToAudio	1.69%
GoToBackpack	1.74%
GoToMarket	2.80%
GoToPlayzone	2.74%
GoToStep	85.03%
GoToTravelJournal	2.87%
Total	100,00%

COUNT OF AI AGENT MESSAGES BY TASK

Current_Task ● DoNothing ● GoToAudio ● GoToBackpack ● GoToMarket ● GoToPlayzone ● GoToStep ● GoToTrave.

Message	Count
Let's explore the World Map!	2.9K
Let's check out the World Map!	2.8K
Let's have a look at the World ...	2.7K
Let's go to the World Map!	2.6K
Should we explore the challen...	2.3K
Hey! Let's do some activities t...	2.1K
Mission time! Choose a challe...	2.1K
Why don't you explore another...	2.0K
Let's have a look at your Trave...	2.0K
My Place sounds fun!	2.0K
My Place time, let's go!	2.0K
Do you want to check your Tra...	2.0K

As shown in Figure 31, most triggered messages relate to the “GoToStep” suggestion, which directs the user to complete the next activity, an indication that the assistant is being used as intended.

Page “Kibot V” investigates whether the number of Kibot activations affects the student’s performance. As we will explore in the next section, no clear correlation was found.

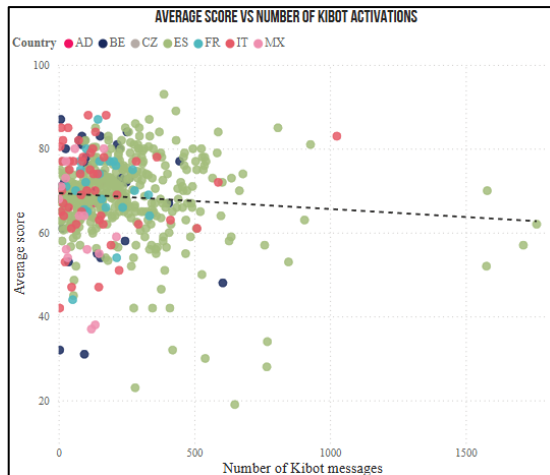


Figure 32: Report page 14 scatter plot of Kibot activations vs grade

The dataset *kibot_with_grades.csv* was generated by merging Kibot events with average performance scores. (code in the *Kibot.ipynb* notebook)

Once the manipulations were completed in Python, visualizing them in Power BI was straightforward.

Finally, “Kibot VI” explores whether students follow the Kibot’s recommendations or not. Specifically, the question is: after receiving a message with a recommended screen (example: GoToMarket), does the student actually navigate to that screen within the next 10 minutes?

A Python function implemented in the *Kibot.ipynb*, was used to calculate this (Figure 33). Findings are discussed further in Section 6.

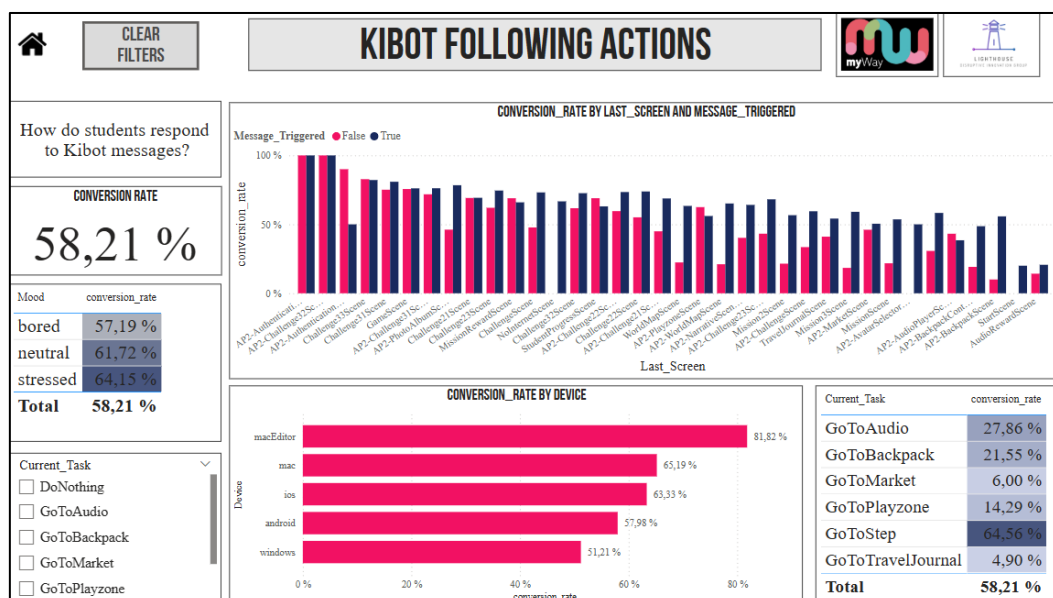


Figure 33: Overview of report page 15

5.4 Predictive Modeling integrated into Power BI

The final methodological step in this project was the development and integration of a predictive model directly into Power BI using Power Query’s scripting capabilities. This model is located in page “Engagement IV” of the Power BI dashboard and complements the rest of the engagement analysis.

The primary objective of the model was to predict whether a given activity was a student’s first (activity.isFirst). This target was selected based on strategic relevance: by identifying students at the start of their journey, educators and platform designers can better assess whether their initial experience correlates with performance and motivation.

5.4.1 Model Design

To identify the most relevant predictors, a correlation heatmap was first generated using Python.

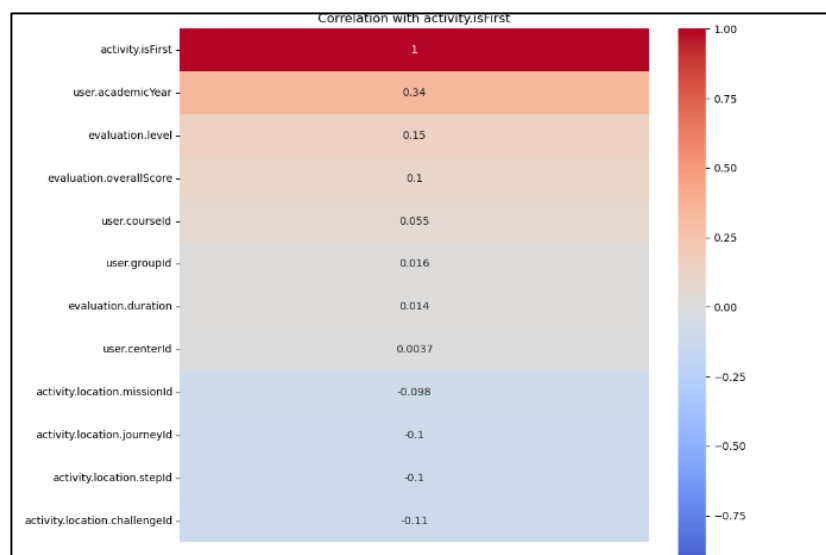


Figure 34: Correlation heatmap of features

Then, two feature sets were tested. The first model included a broader selection of variables. The full heatmap is in Appendix A: Additional Figures, Figure 61. The second model focused on a smaller set of predictors selected from Figure 34 above.

Random Forest classifier was used for the model, due to its robustness and ability to handle mixed data types without intensive preprocessing. (Breiman, 2001) Hyperparameters were kept light ($n_estimators=20$, $max_depth=3$) to ensure a fast execution within Power BI’s Python runtime.

After comparing both models, the result of which comparison is in the appendix (Table 8: Table of Models Comparison), the simplified version (Model 2) achieved higher accuracy and was faster to compute, so it was selected as the final version.

5.4.2 Implementation in Power BI & utility

The selected model was built directly in Power Query using the “Run Python Script” block (Figure 35). To ensure smooth performance, especially given runtime and memory constraints, a random sample of 5,000 activities was used for model training. The model input was a clean table prepared earlier in Power BI’s transformation layer, and all processing steps until prediction are performed inside the same block of code within Power Query.

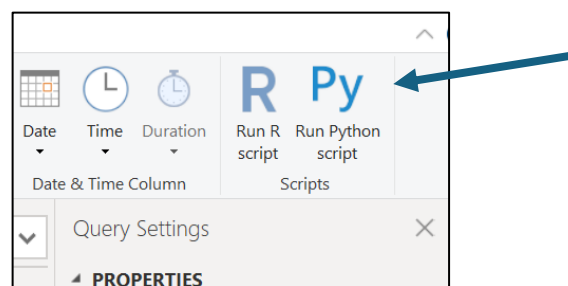


Figure 35: Run Python button in Power Query

Within the script, a train-test split was performed, producing a classification report with accuracy, precision and recall. Additionally, feature importance was extracted from the trained model and visualized using a separate plot in Power BI.

The model’s predictions were returned as a simple binary column (prediction) appended to the main dataset. This format enables an easy integration with visuals.

Figure 36 shows how the model output is presented in the “Engagement IV” page. A key design element is the metric selector, where the user can decide between visualizing prediction labels by duration or by score. This deepens the analytical capabilities of the report.

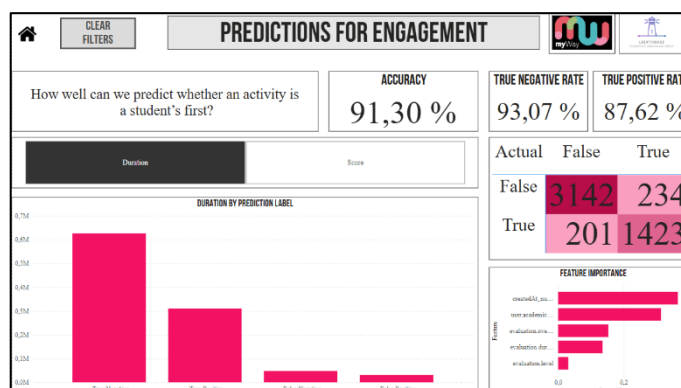


Figure 36: Overview of report page 9

With this integration, the dashboard allows Lighthouse and Kids&Us to not only analyze behavior retrospectively, but also see student trajectories in real time, especially during the first stages of student interaction.

For a discussion on practical and technical constraints encountered during this process, please refer to the following section: Limitations 7.3.

6. Results

This section presents the main results obtained from the analysis of the MyWay platform data. These findings are organized around the four central pillars explored throughout the project.

6.1 Usability Findings

Usability analysis helped understand how students use the MyWay platform and whether there are areas of the platform that create confusion.

One of the key features was the interactive click heatmap (Figure 17) highlighted in section 5.3.3. This helped identify which areas received the most interaction. As expected, highly interactive pages such as GameScene or WorldMapScene screens showed dense clicking patterns. As for the Challenge screens (activities), the density of clicking drops rapidly from Challenges 1 and 2, to Challenges 3 and 4. (Figure 37, Figure 38) While this may be due to the timeliness of the data, suggesting students may not have progressed that far, it raises the possibility of drop-off or disengagement. Further analysis is recommended to determine if these patterns are natural.

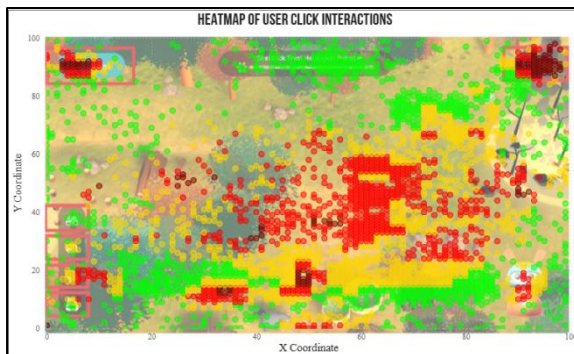


Figure 37: Click interactions Mission 3 Challenge 1

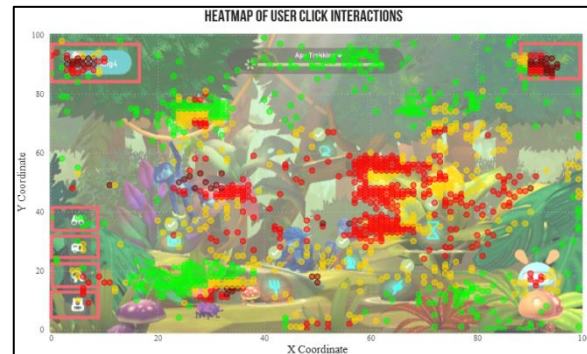


Figure 38: Click interactions Mission 2 Challenge 1

Furthermore, as seen in both figures above, a large number of users frequently clicked on the Kibot assistant icon when no direct interaction was expected. This interaction suggests that

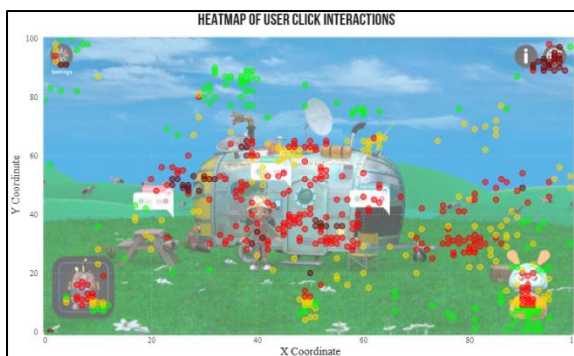


Figure 39: Click interactions Mission 1

users might expect the Kibot to be interactive or are looking for more guidance in specific moments. This opens the door to developing a “Kibot On-Demand” feature, allowing the user to call on the assistant when needed, rather than waiting for automated triggers.

A closer look at the clicks labeled as very high density, revealed a lot of clicks on the AudioPlayerScene, especially in the top-right corner, where there is a “close” icon. This suggests frustration or rushed exits. When analyzed individually, almost all students showed at least one high-density cluster in this area, which could be flagged for UX review.

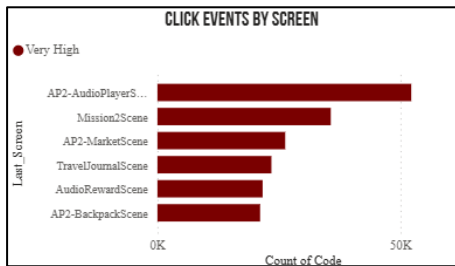


Figure 40: Screens with most "very high" density clicks

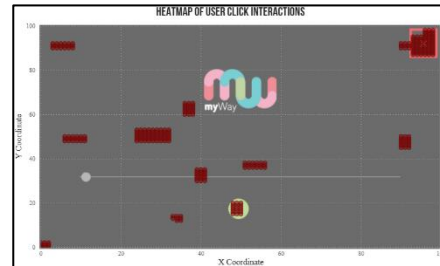


Figure 41: AudioPlayerScreen Click interactions

Another insightful finding Navigation behavior was visualized through a Sankey diagram (Figure 20). Although the unfiltered diagram appears messy due to data volume (fig), it automatically shows expected behavior: a lot of actions from WorldMapScene ending in MissionScene, GameScene or AudioPlayerScene.

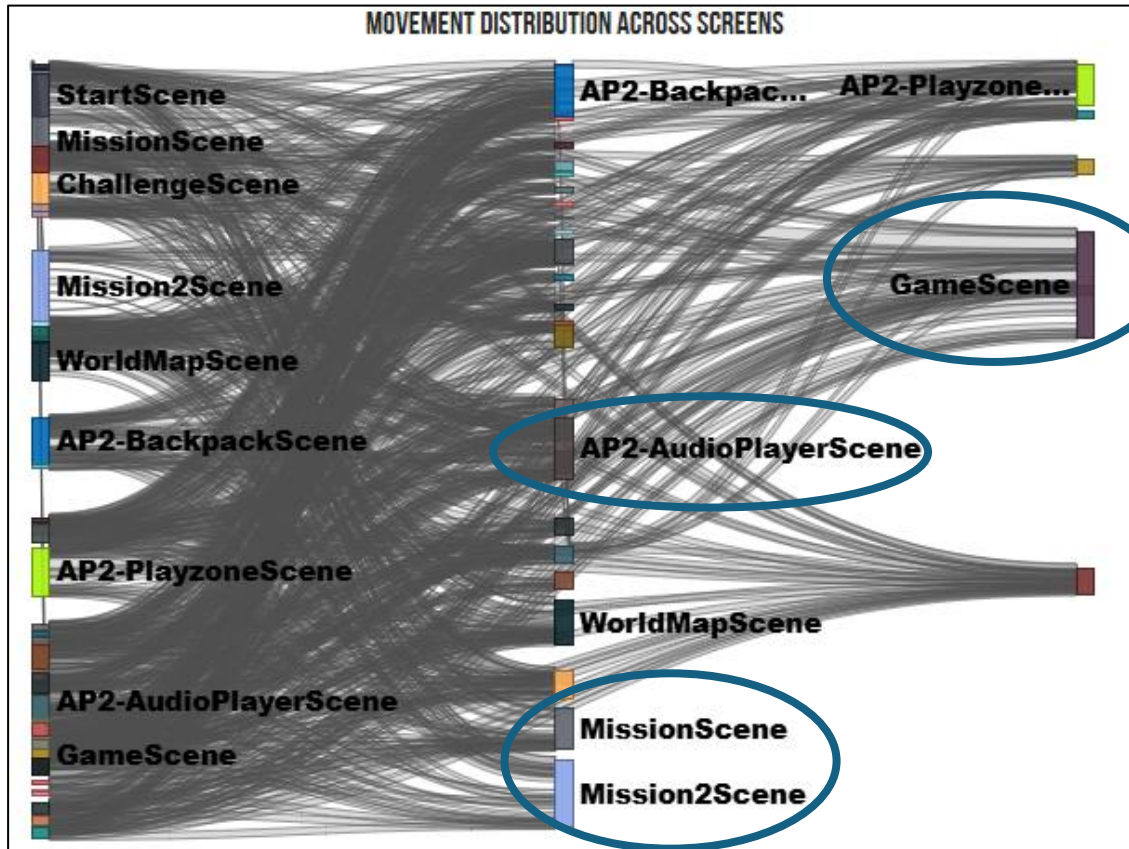


Figure 42: Unfiltered Sankey Diagram

Filtered diagrams provided a cleaner breakdown highlighting movements. Filters by destination screen (Figure 43), show interactions that led to AudioPlayerScene. Interestingly, many originate from the PlayzoneScene, which may suggest students seek a familiar activity like listening to an audio.

Figure 44 focuses on origin screens, specifically WorldMapScene. The majority of students proceed to MissionScenes, confirming these are well integrated in the platform's progression. A smaller portion navigate to BackpackScene, which may be a sign of interest in the backpack content, or doubt before doing a task.

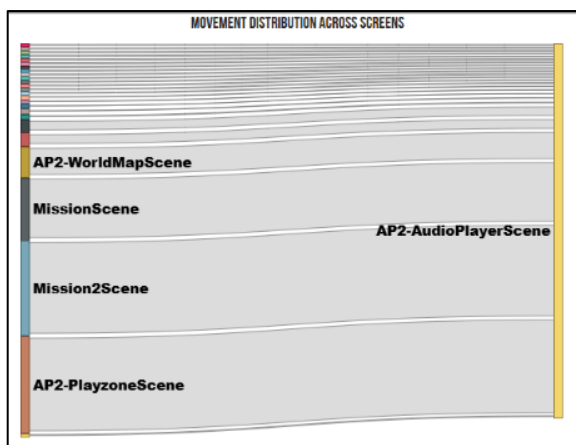


Figure 43: Sankey Diagram filtered by destination

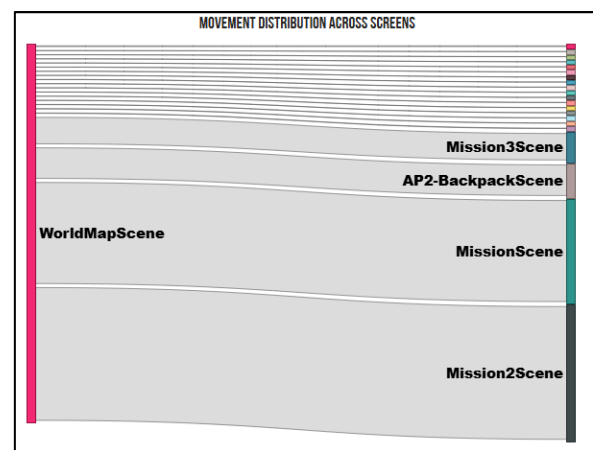


Figure 44: Sankey Diagram filtered by origin

Finally Figure 45 shows navigation by individual user. In this specific case the student begins at StartScene, alternates between WorldMapScene and BackpackScene and moves into Challenge and Game screens. This individual-level view provides context for identifying irregular navigations.

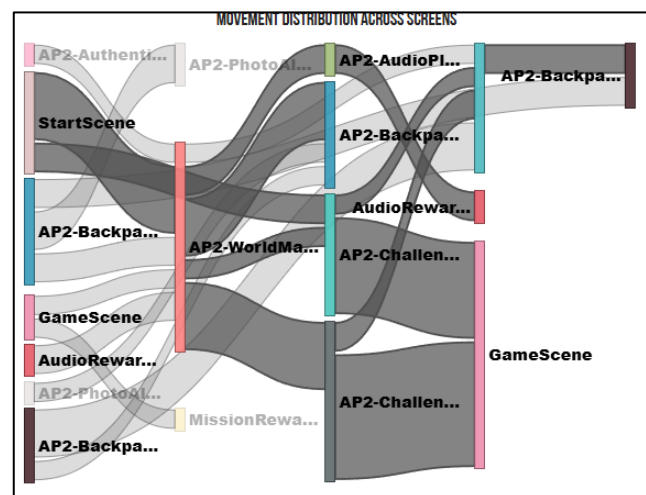
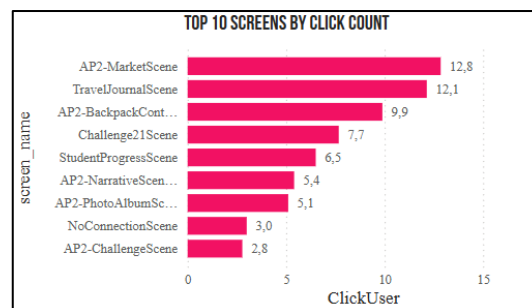


Figure 45: Sankey Diagram filtered by User ID

Clustering users by click frequency and time spent revealed five distinct behavioral profiles, as seen in Figure 23: Scatter plot with clusters report page 4. Balanced users showed consistent interaction on learning-focused screens like GameScene and AudioPlayerScene, suggesting an effective usage flow. Confused Observers, in contrast, spent more time but clicked less which may lead to confusion.

Over-Clickers showed high activity in screens such as MarketScene and JournalScene, which could signal frustration or misuse of the platform (Figure 46).

Figure 46: Top 10 clicked screens for Over-clickers



6.2 Engagement Trends

The engagement analysis reveals several insights about how students interact with the MyWay platform. A key engagement trend is the overuse of a specific playzone activity (Figure 47).

One user repeated it 700 times (Figure 63 Appendix A: Additional Figures). This suggests either a bug where the activity is triggered repeatedly, or that students are intentionally returning to it. If the latter is true, it represents an important opportunity: identifying this activity's mechanics could increase engagement across the platform.

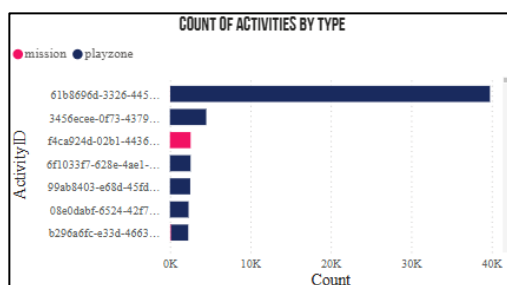


Figure 47: Count of activities

In terms of performance, most students achieve relatively high scores, with an average grade of 67.08. However, when analyzing grades by skill, writing and speaking show lower performance levels, likely reflecting higher difficulty or lower support for those tasks on the

platform, pointing areas to add tools and guidance.

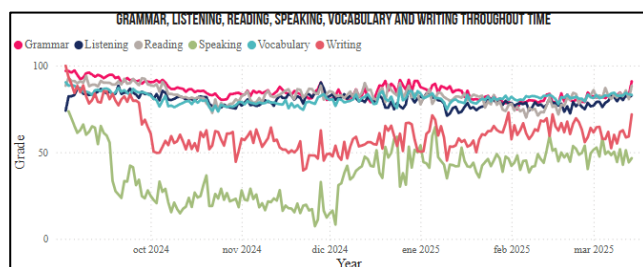


Figure 48: Line chart of skill grades throughout time

Country-level analysis shows that while Spain represents the majority of users, average scores differ significantly across regions. For instance, students in Czechia and Andorra consistently perform above average, whereas Mexico and France show lower mean scores, possibly indicating regional disparities in curriculum alignment or user experience. However, the samples of other countries are considerably smaller.

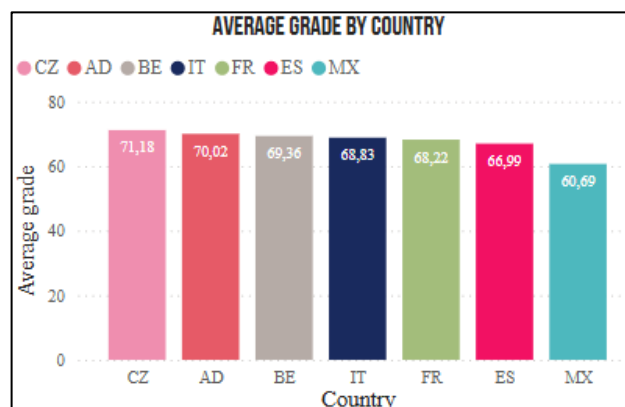


Figure 49: Average score per country

Finally, the box plot (Figure 26: Report page 8 boxplot) confirms that mid-range durations (60 to 300 seconds) tend to result in higher performance, supporting the idea that longer durations could indicate distraction or difficulty, and do not necessarily result in higher grades. This suggests a potential value in adding time guidance triggers related to the Kibot.

6.3 Kibot Impact Analysis

The Kibot assistant was one of the central features under investigation and the goal was to assess its effect on learning.

Initial analysis (“Kibot I” page) revealed that Kibot messages were triggered regularly, particularly messages tied to the “GoToStep” anchor, which encourages students to proceed to an activity. This behavior was consistent with the platform’s intended pedagogical guidance. (Figure 28: Report page 10 bar chart displaying frequency and trigger rates of AI Agent)

However, when assessing Kibot’s impact on academic performance (“Kibot V” page), the results were neutral. No significant correlation was found between the number of Kibot activations and overall grades (Figure 32: Report page 14 scatter plot of Kibot activations vs grade). This suggests that while Kibot may help with platform navigation and task initiation, it does not directly influence academic performance. However, if data is analyzed by CenterID, we do find cases where a relation can be seen (Figure 50).

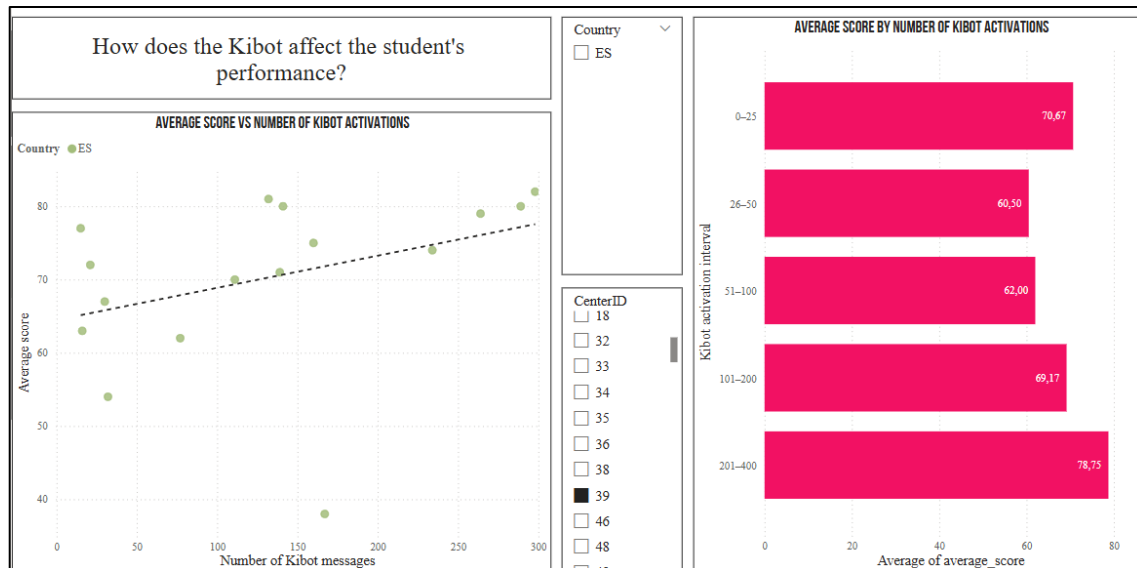


Figure 50: Kibot-grade analysis for center ID 39

Conversion rate analysis, measuring whether a student followed the Kibot’s recommendation within 10 minutes, was more promising. The average conversion rate was found to be 58.21% for all Kibot messages (“Kibot VI” page). More than half of Kibot messages successfully guided students to the intended destination.

Current_Task	conversion_rate
GoToAudio	27,86 %
GoToBackpack	21,55 %
GoToMarket	6,00 %
GoToPlayzone	14,29 %
GoToStep	64,56 %
GoToTravelJournal	4,90 %
Total	58,21 %

However, the effectiveness varied greatly by anchor type, with “GoToStep” having the highest success rate, and all the rest very low. This means that students are responsive to actionable messages but may ignore messages perceived as less relevant.

Figure 51: Conversion rate per estimated task

This presents a design challenge: the Kibot must evolve into a more intelligent guide capable of providing adaptive and contextually relevant suggestions.

6.4 Predictive Model Outcomes

The final component of the analysis was a predictive model aimed at identifying whether a given activity was a student’s first interaction. The goal was to better understand onboarding trends and identify patterns at an early stage of usage.

The model's results were visualized using a confusion matrix, bar charts segmented by prediction type (TP, TN, FP, FN) (Figure 14). True positives and False positives had the highest average scores, suggesting that early activities are associated with high engagement and motivation. In terms of duration, True positives and False negatives took longer, supporting the idea that first-time users explore more slowly, while returning users move quicker.

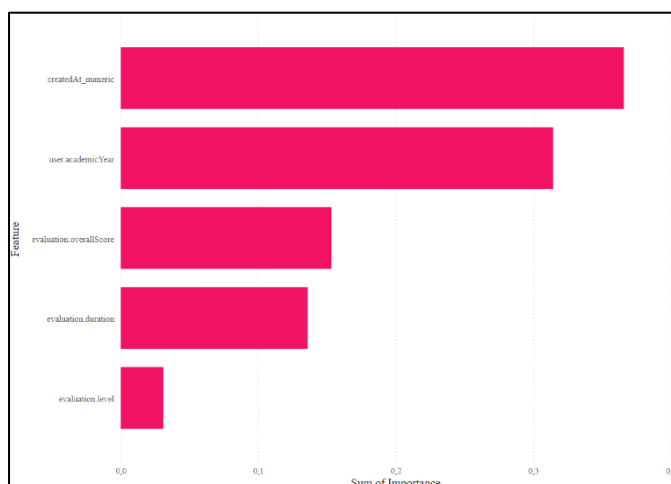


Figure 52: Feature importances

However, when examining feature importance, it became clear that the most influential predictors were date and academic year.

This points out the risk of temporary bias. In future models a stricter feature selection should be applied to avoid this.

In this current form, the model serves more as a detector of when a student started using the platform, rather than uncovering deep patterns that define first-time users. Given the structure of the dataset and the high proportion of missing or null values, the model had to rely on a reduced set of usable variables. While this can be useful as well, to evaluate the initial idea, a more rigorous approach could be implemented with a complete dataset.

From a practical perspective, identifying students at the start of their journey opens the possibility of tailoring experiences, adapting to difficulty levels, or with motivational messages.

Even with its limitations, this model demonstrates how lightweight machine learning can be embedded within Power BI to enrich dashboards.

7. Discussion and Conclusions

7.1 Recap of results

The idea of the project was to evaluate engagement and usability on the MyWay learning platform.

As for usability, click heatmaps and navigation flows revealed clear patterns: certain screens like GameScene and WorldMapScene acted as central interaction screens, while others, such as last Challenges or AudioPlayerScene, suggested drop-off or confusion. The high click density on close buttons or Kibot icons implied potential frustration, or unmet expectations for support, insights valuable for future design improvements.

Engagement analysis added another layer of insight. While average performance was strong, temporal patterns showed a post-launch decline in activity, common in digital learning but indicative of the need for long-term motivation strategies. Analysis by specific kills showed weaker results in writing and speaking, most likely due to their higher difficulty and lower system support. In contrast, the overuse of a specific playzone activity, repeated over 700 times by one user, pointed either to design flaws or a highly engaging activity, still in need of more analysis.

Kibot usage data revealed that while the assistant is frequently triggered (especially via the "GoToStep" message), it doesn't directly correlate with improved academic performance. However, it does show strong guidance utility: over 58% of its messages successfully led students to the intended screen. Still, its effectiveness is only toward activity related messages, suggesting the need to evolve its relevance.

Finally, the predictive model, integrated into Power BI, successfully identified first-time student activities. These "firsts" were linked with higher scores and longer durations, suggesting that onboarding moments are critical touchpoints in the student experience.

7.2 Value of Power BI

The core contribution of this project was the implementation of a comprehensive Power BI dashboard. Power BI proved to be an effective tool for synthesizing large-scale education data. The ability to embed Python scripts enabled advanced statistical analyses, while the visual storytelling components (click heatmaps, Sankey diagrams, clustered scatter plots, and tooltips) allowed stakeholders to intuitively see patterns.

7.3 Limitations

While this project provides an analytical overview of student engagement, usability, and AI-assisted interactions on the MyWay platform, several limitations affected both the technical implementation and the analytical depth of the insights.

These limitations are important to consider for future iterations or when interpreting the results presented in Section 6 above.

- Performance and Integration Constraints

The initial goal of the project was to build a fully integrated tool within Power BI, where all Python scripts were directly executed in Power Query to keep all transformation and modeling unified. However, some Python functions, particularly those for Kibot conversion analysis, required row-by-row matching, leading to execution times exceeding several hours. Power BI's integration with Python via Power Query is not optimized for long-running scripts.

As a result, several advanced transformations were precomputed in Python notebooks, and their output (*kibot_with_grades.csv*, *screen_summary.csv*, *repeated_actions_summary.csv*) was manually imported into Power BI. While this ensures performance and reliability within the dashboard, it introduces some disconnection from the fully dynamic vision initially intended.

- Sampling

Sampling was applied to ensure responsiveness and limit large computations. The predictive model was trained on a sample of 5,000 activities (out of 200,000) and in conversion rate calculations for Kibot recommendations, only the first 100 Kibot messages were taken.

These sampling decisions were necessary for responsiveness and usability, but they also mean that results are indicative, and there may be edge patterns not captured in the sample.

- Data Quality and Structure

As for the structure and quality of the data itself, “Mood” was heavily skewed, with over 75% of events labeled as “bored” and less than 1% as “engaged” or “frustrated”. This made mood a poor candidate deeper analysis. Also, the Activity Reports initially included only a sample of 10 anonymized users. The full 1,600-user dataset was made available at a later stage, requiring adaptations midway through the project. Finally, the timestamps and time zone consistency across datasets required normalization and may still have some errors in specific cases.

7.4 Recommendations and Future work

Based on the findings, several recommendations are proposed for both platform development and future research:

- Enhance UX for some screens: Screens like AudioPlayerScene and Challenges 3-4 should have UX reviews. Adding onboarding messages, clearer navigation, or in-task motivation prompts may reduce disengagement.
- Develop a Kibot on-demand function: Given students' tendency to click the Kibot even without prompts, allowing users to “call” the assistant could shorten the gap between expected and actual interactivity. This would increase perceived support and autonomy.
- Make Kibot messages smarter: Beyond “GoToStep”, other message types did not perform well. Improving message contextualization would likely increase follow-through rates.
- Support low-scoring skills: Writing and speaking activities require more guidance, such as feedback loops, audio guides, or peer examples. These additions could help improve learning outcomes in these difficult areas.
- Monitor repeat behavior: The case of the overused playzone activity should have a deeper analysis. If it is due to a bug, it should be fixed, if it reflects genuine interest, its design should be studied and replicated.
- Use predictive modeling to personalize onboarding: The early activities of students are crucial. The model should be used to implement adaptive journeys, offering simplified tasks, motivational messages, or tailored paths for new users.
- Iterate on Power BI integration: While the dashboard has strong functionality, future versions could explore end-to-end dynamic modeling using lightweight scripts or integrating with specialized platforms for live prediction updates.

In conclusion, this project has demonstrated the power of data visualization and machine learning for understanding and improving digital learning experiences. By combining detailed interaction analysis with predictive insights and making these findings accessible through Power BI, the project not only uncovers current gaps but lays the foundation for a smarter, more responsive learning platform.

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9. Appendix

9.1 Appendix A: Additional Figures

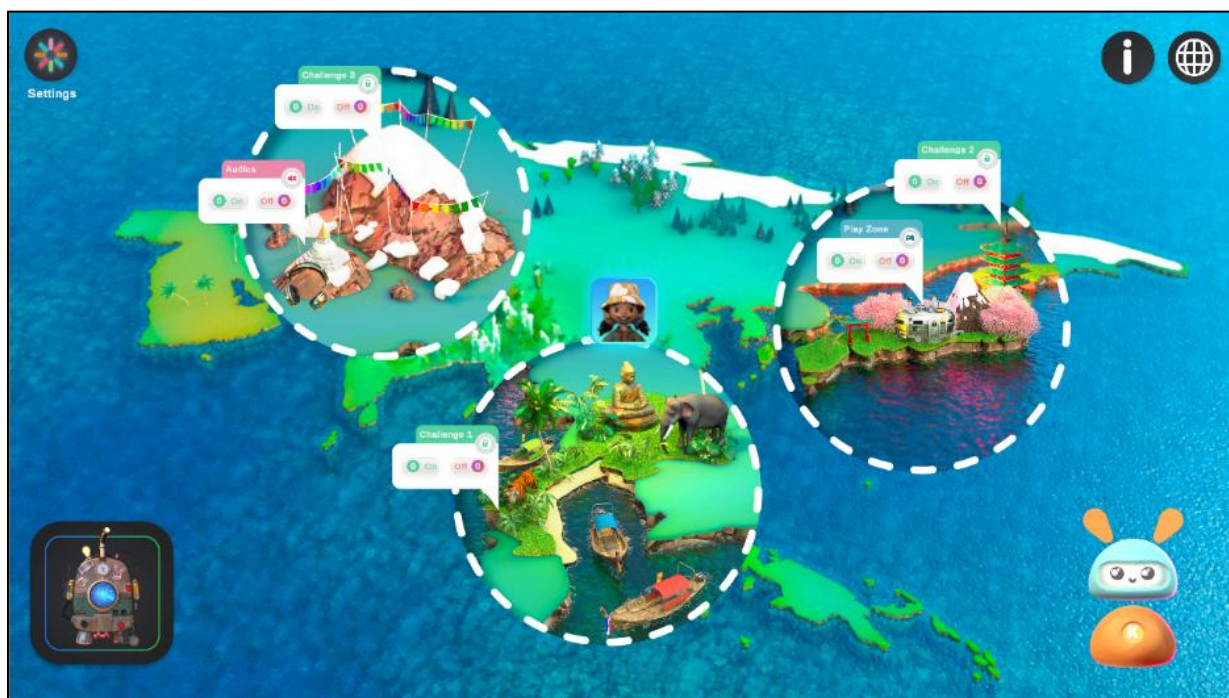


Figure 53: MyWay mission overview

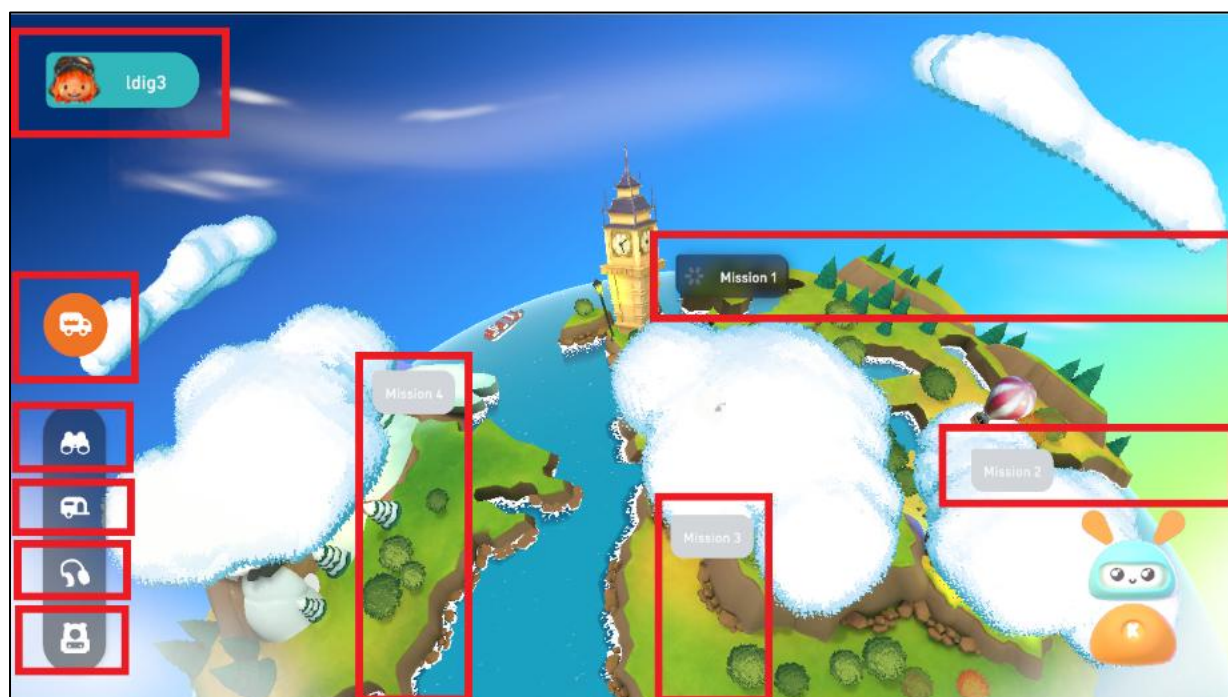


Figure 54: MyWay world map overview

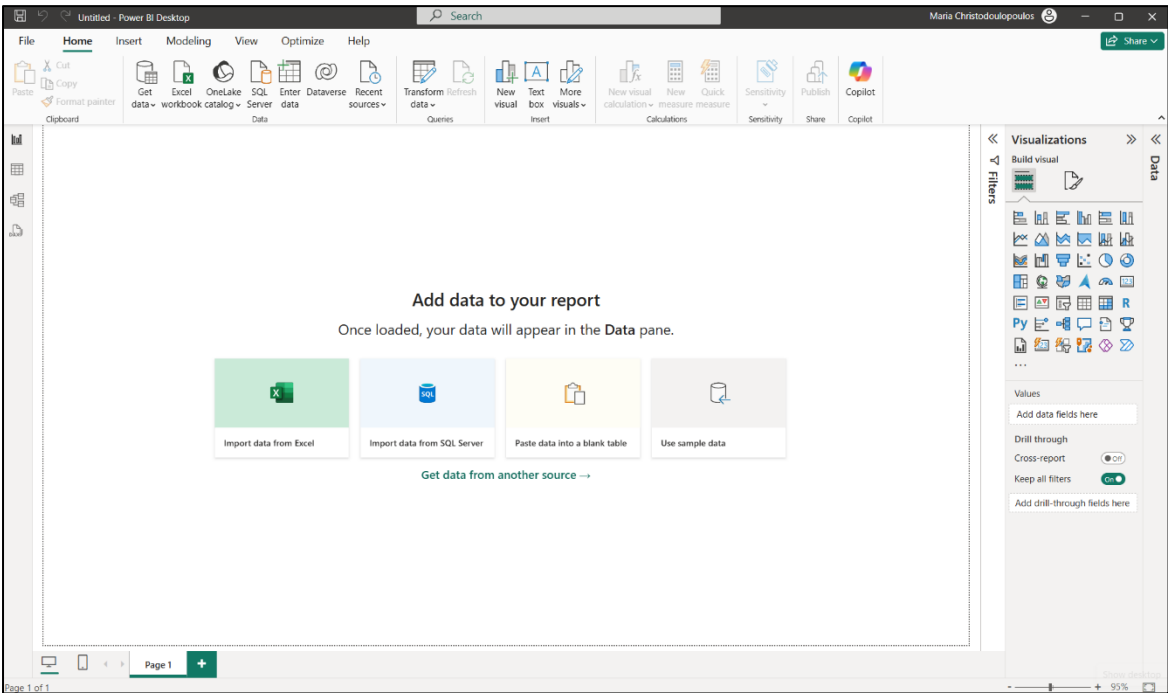


Figure 55: Power BI overview of opening page

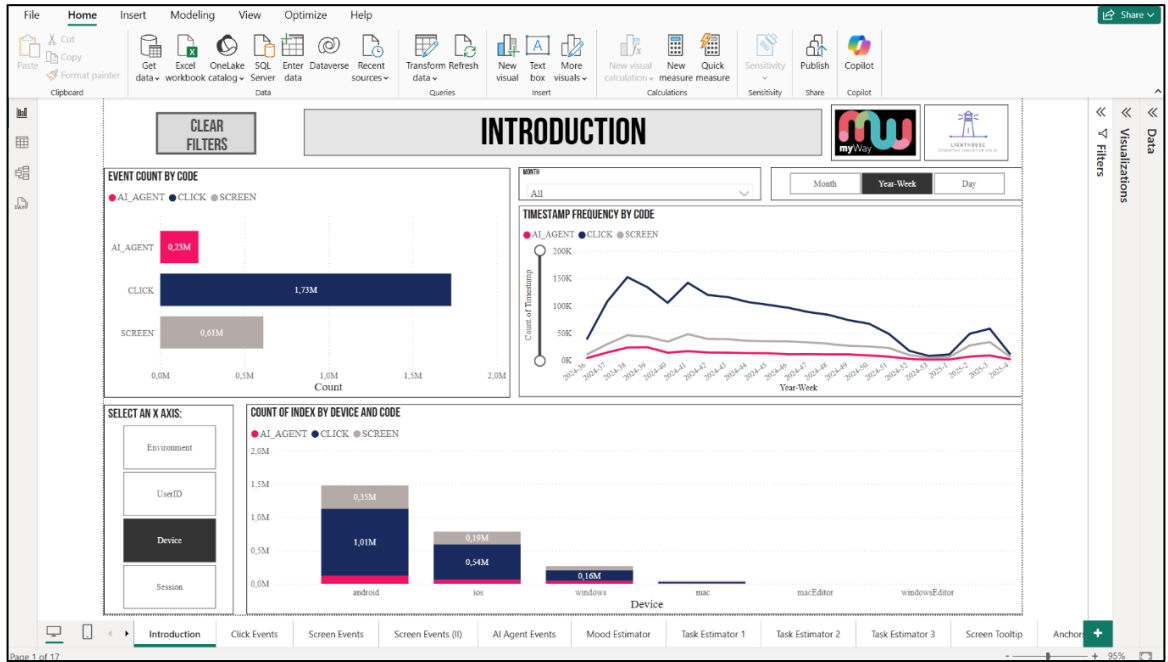


Figure 56: Example of a page of a Power BI report

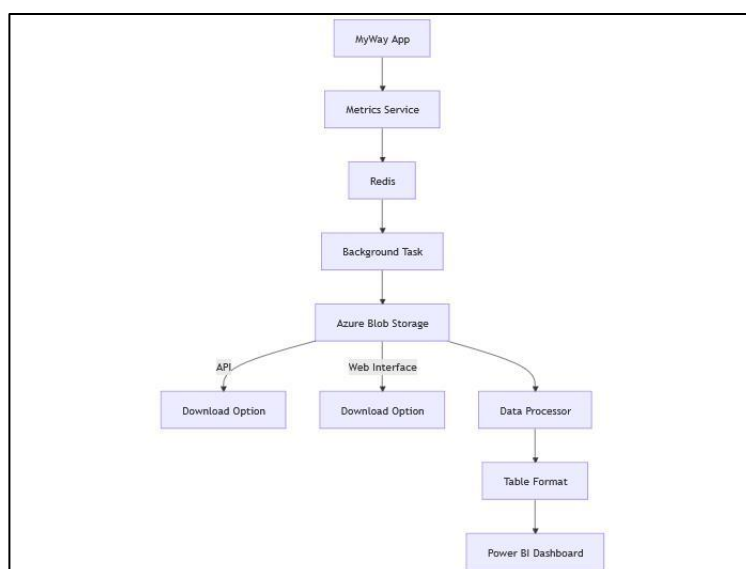


Figure 57: Data acquisition pipeline

_2XA9UDj46vyaj5PgVmvIw==	✓	08/04/2025 12:07
_4LmJMxuwka-d2ao6fUAPw==	✓	08/04/2025 12:07
_7_xgxRxFGyVcotfDacMw==	✓	08/04/2025 12:07
7n0B-oniQbilP96X1kZA==	✓	08/04/2025 12:07
_9o1Ztl0whx6x3SxMcuz3w==	✓	08/04/2025 12:07
_17APtZdbJ0DfKloSyUV-g==	✓	08/04/2025 12:07
_Anm8icZtklOjwfe7mY_Q==	✓	08/04/2025 12:07
_Bh7K5Lw4Aw_XWxTIMIX6Q==	✓	08/04/2025 12:07
_bNSujG8AvDVtx3EyJSjQ==	✓	08/04/2025 12:07
_bWERxdNOjalgfN9ryoGsQ==	✓	08/04/2025 12:07
_cmDuEAbMJefGwX3orn2qA==	✓	08/04/2025 12:07
_DBdwbPPvfiJwKw0ThQjg==	✓	08/04/2025 12:07
_djQ-sQ1wU3Da2odK77CRA==	✓	08/04/2025 12:07

Figure 58: User ID individual folders with interaction data

1726507184.json	✓	21/01/2025 11:22
1726508445.json	✓	21/01/2025 11:22
1726599557.json	✓	21/01/2025 11:22
1726600455.json	✓	21/01/2025 11:22
1727712388.json	✓	21/01/2025 11:22
1730385374.json	✓	21/01/2025 11:22
1730386275.json	✓	21/01/2025 11:22
1730387177.json	✓	21/01/2025 11:22
1733156017.json	✓	21/01/2025 11:22
1733156918.json	✓	21/01/2025 11:22
1733157898.json	✓	21/01/2025 11:22

Figure 59: Timestamp compressed JSON folders inside a User ID folder

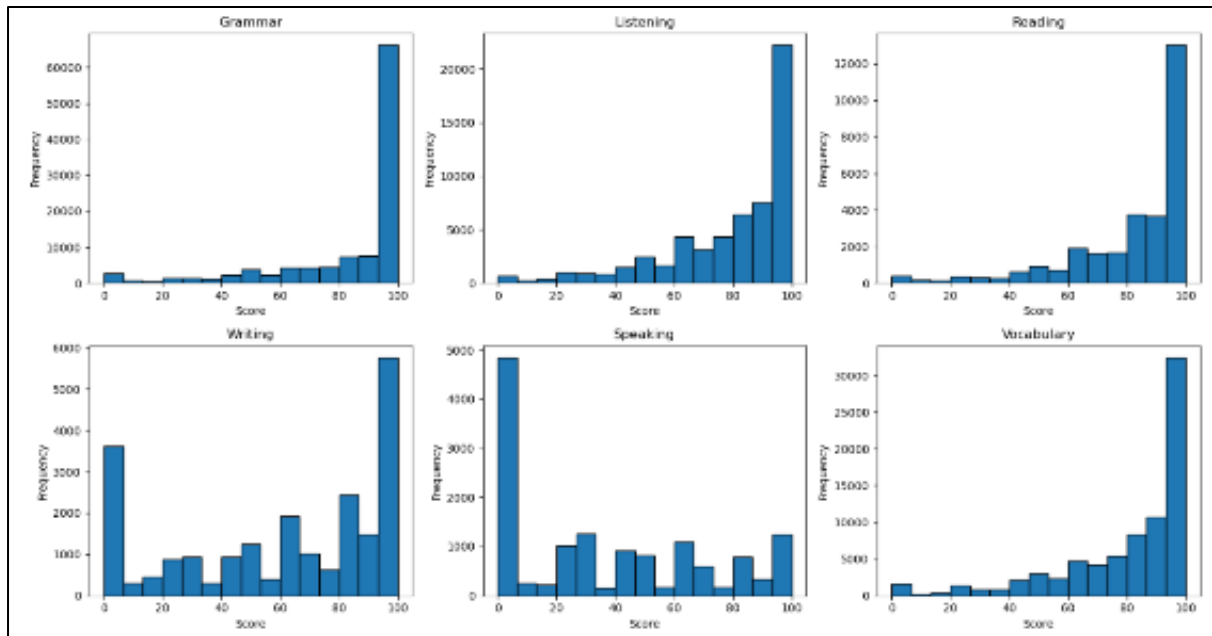


Figure 60: Language subcategories score distribution

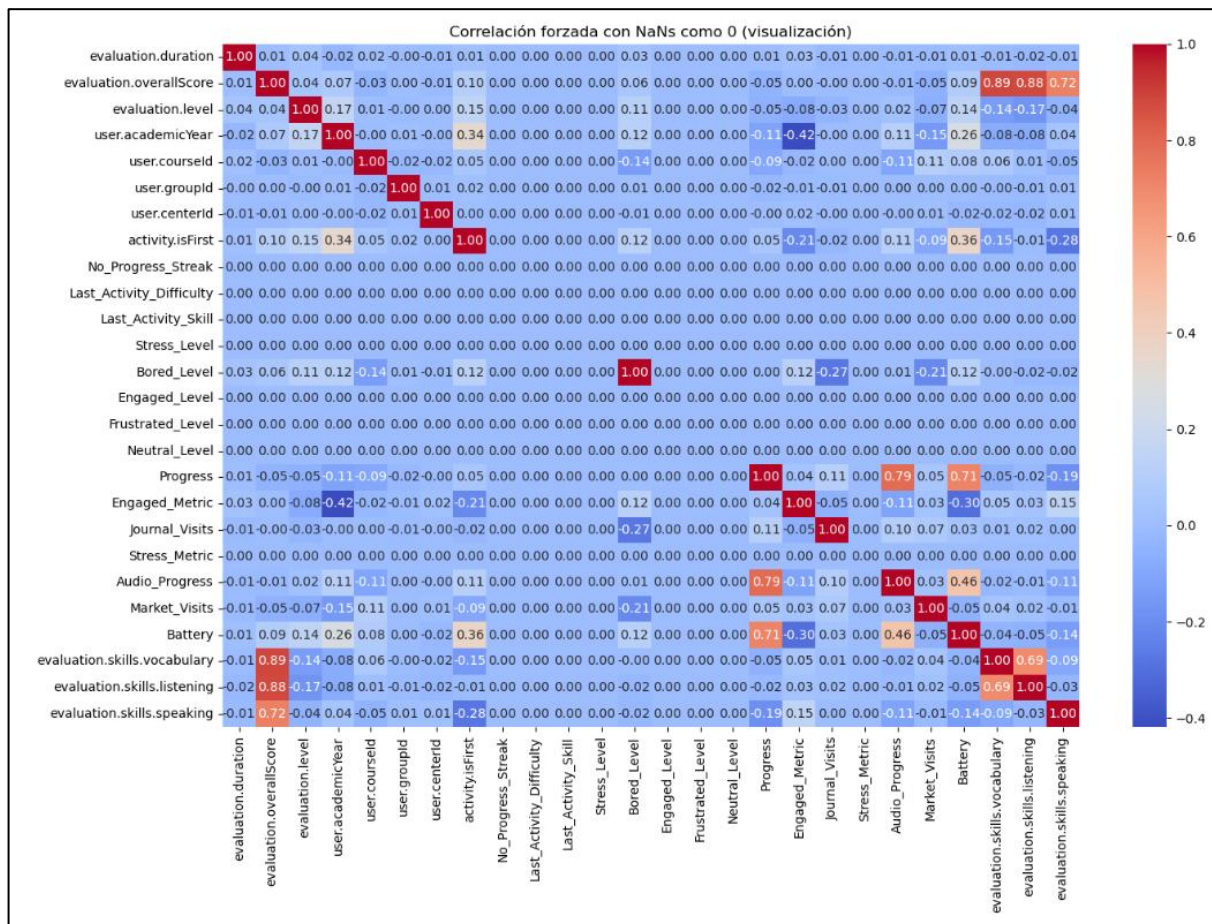


Figure 61: Correlation heatmap of all variables

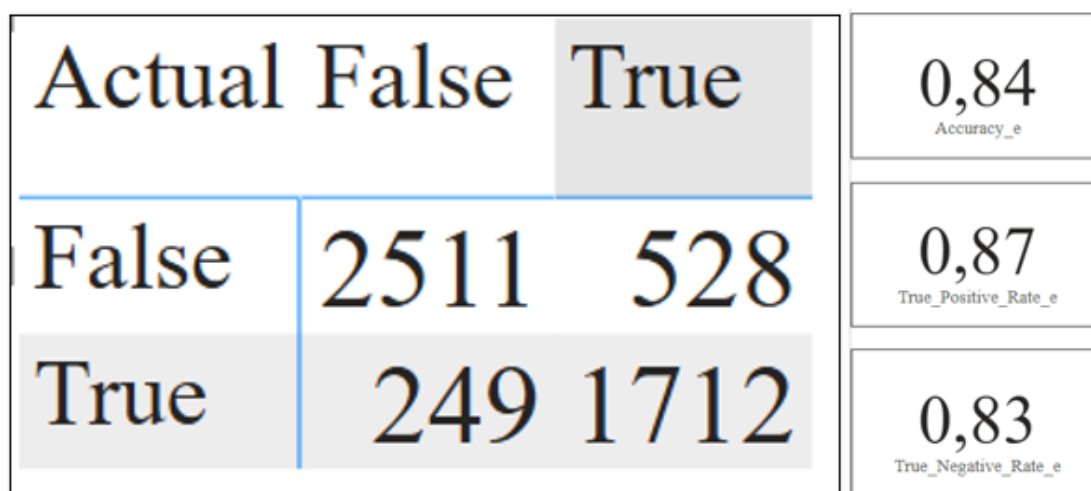


Figure 62: Model 1 results

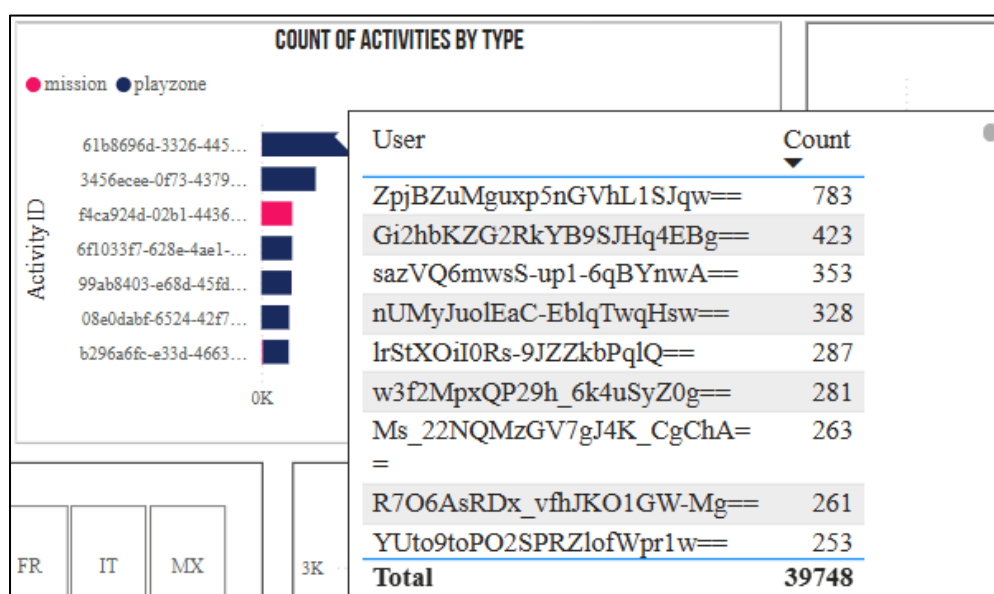


Figure 63: Tooltip of activity count per user

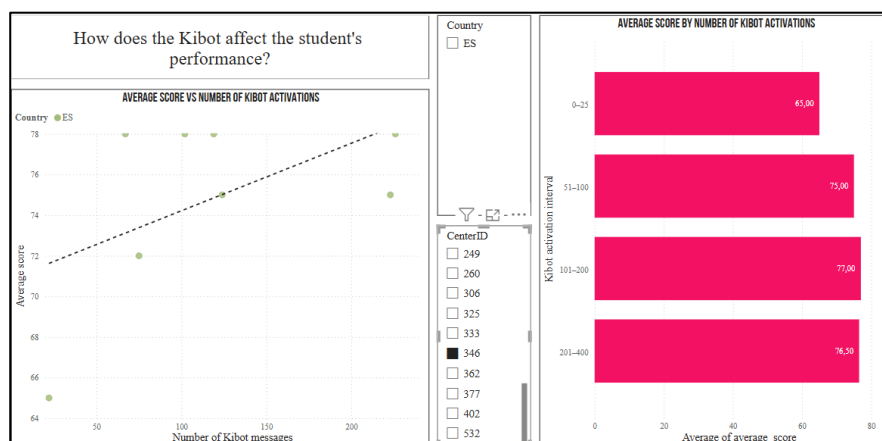


Figure 64: Example 2 of Center ID kibot-grades analysis

9.2 Appendix B: Additional Tables

Table 5: Variables in metrics.csv

VARIABLE	DESCRIPTION
Mood	Estimated mood of the student during the event.
Previous_Task	The previous task completed by the student.
Current_Task	The task estimated to be performed next.
AI_Agent_Message	Whether an AI Agent message was sent.
No_Progress_Streak	Number of events without progress.
Last_Activity_Difficulty	Difficulty level of the last completed activity.
Last_Activity_Skill	Skill type of the last completed activity.
Stress_Level	Level of stress detected (scale 0-1).
Bored_Level	Level of boredom detected (scale 0-1).
Engaged_Level	Level of engagement detected (scale 0-1).
Frustrated_Level	Level of frustration detected (scale 0-1).
Neutral_Level	Level of neutrality detected (scale 0-1).
Progress	Progress made in the current task or session.
Engaged_Metric	Calculated engagement metric (custom logic).
Journal_Visits	Number of visits to the Journal screen.
Stress_Metric	Calculated stress metric (custom logic).
Audio_Progress	Progress made in audio-related tasks.
Market_Visits	Number of visits to the Market screen.
Battery	Battery level or status of the device.
Event_ID	Unique identifier for the event.
Timestamp	Timestamp of the event.
SessionID	Unique session identifier.
OrchestratorTimestamp	Timestamp from the orchestration system.
UserID	Anonymized unique student ID.
Device	Type or model of the device used.
Environment	Environment or platform used (e.g., browser, OS).
Type	General type of event.
Code	Specific event code (e.g., Click, Screen, AI Agent).
Message_X	X coordinate of click events.
Message_Y	Y coordinate of click events.
Message_Type	Type of the message triggered.
Message_From	Screen from which the user navigated.
Message_To	Screen to which the user navigated.
Message_Mood	Mood associated with the triggered message.
Message_Anchor	Anchor or trigger type of the Kibot message.
Message_Branches_Value	Value assigned to branching logic of message.
Message_Triggered	Whether the message was actually triggered.
Message_User_Input	Whether the user responded to the message.
Message_AI_Action	Action taken by the AI assistant.
Last_Screen	Screen where the interaction occurred.

Table 6: Variables in activityreports.csv

VARIABLE	DESCRIPTION
_id	Unique identifier for the database entry.
user.id	Anonymized unique identifier for each student.
user.courseId	Identifier for the course in which the student is enrolled.
user.academicYear	Academic year associated with the student.
user.groupId	Group or class identifier for the student.
user.centerId	Identifier of the center or school the student belongs to.
user.country	Country associated with the student.
activity.id	Unique identifier for the activity performed.
activity.type	Type of activity: e.g., "mission" or "playzone".
activity.isFirst	Boolean indicating whether this is the student's first activity (True/False).
evaluation.overallScore	Final score given for the activity.
evaluation.level	Difficulty level of the activity.
evaluation.duration	Time taken to complete the activity, in seconds.
timeZone	Time zone in which the activity was completed.
createdAt	Timestamp of when the activity was submitted.
updatedAt	Timestamp of the most recent update to the record.
__v	Version key used internally by the database system (MongoDB).
evaluation.skills.vocabulary	Final vocabulary score for the entire activity.
evaluation.skills.grammar	Final grammar score for the entire activity.
activity.location.missionId	Identifier for the mission associated with the activity.
activity.location.challengeId	Identifier for the challenge associated with the activity.
activity.location.journeyId	Identifier for the broader journey or unit the activity belongs to.
activity.location.stepId	Identifier for the specific step within the mission or challenge.
evaluation.skills.listening	Final listening score for the activity.
evaluation.skills.reading	Final reading score for the activity.
evaluation.skills.writing	Final writing score for the activity.
evaluation.skills.speaking	Final speaking score for the activity.

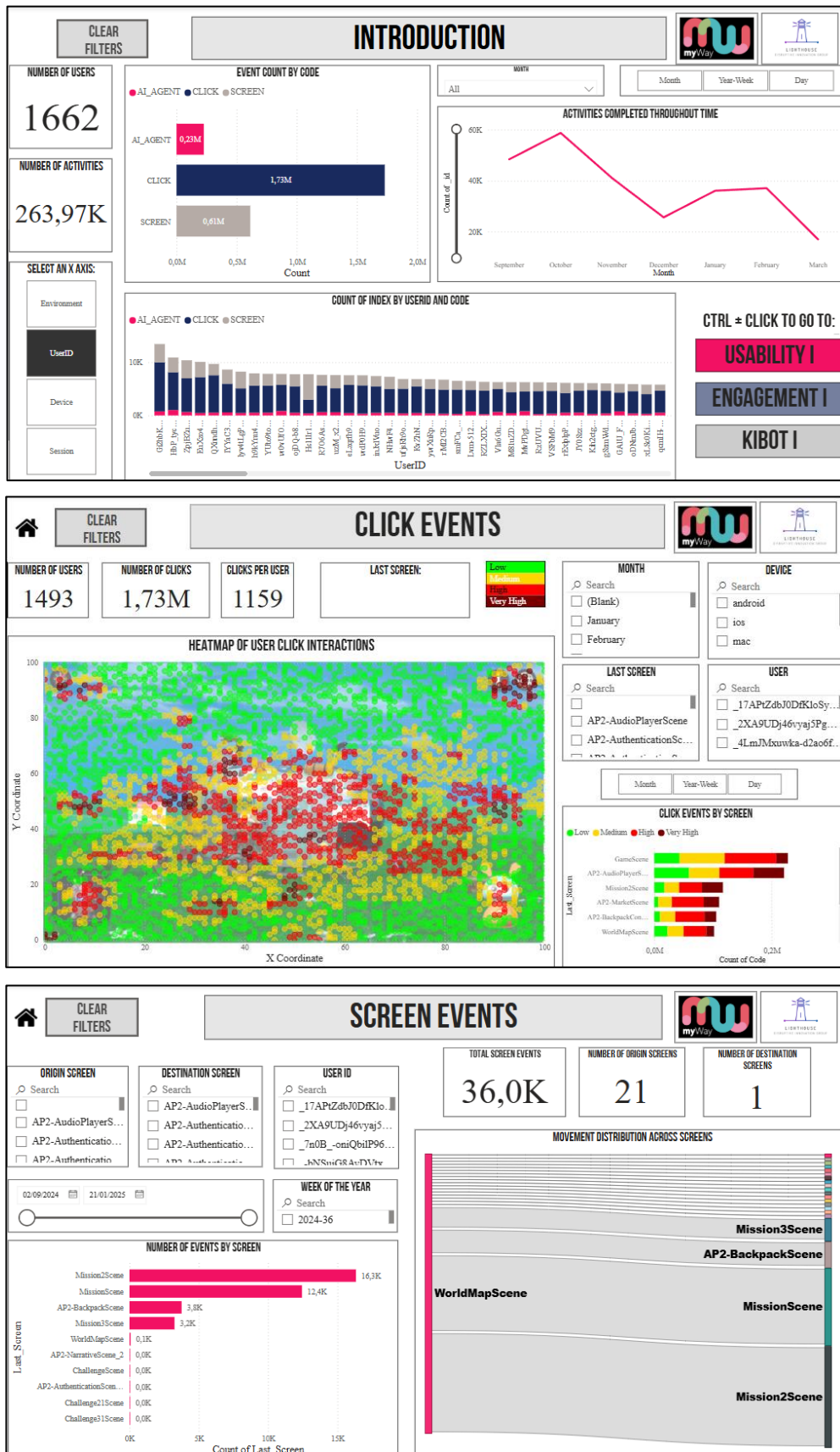
Table 7: Summary of data sources

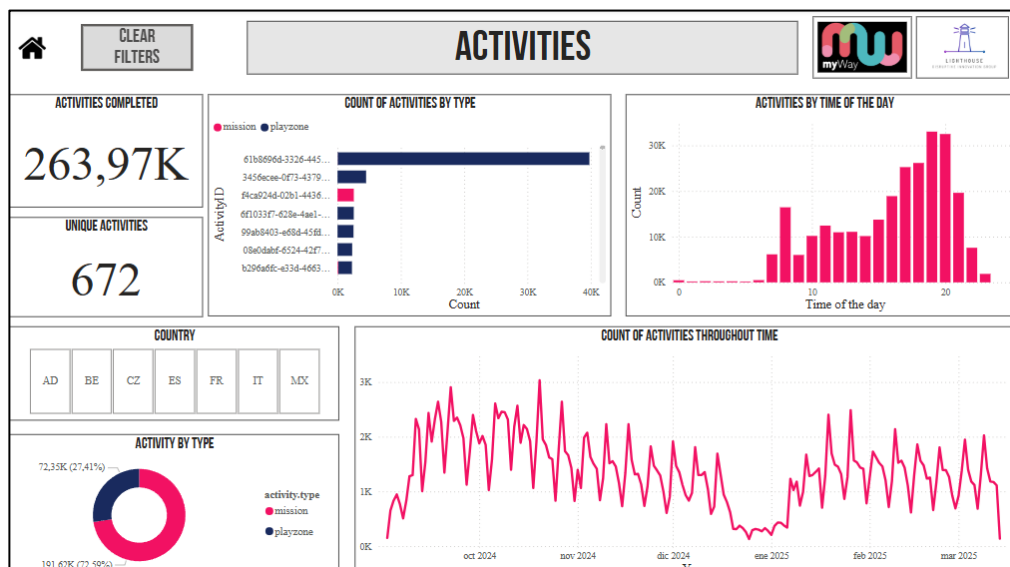
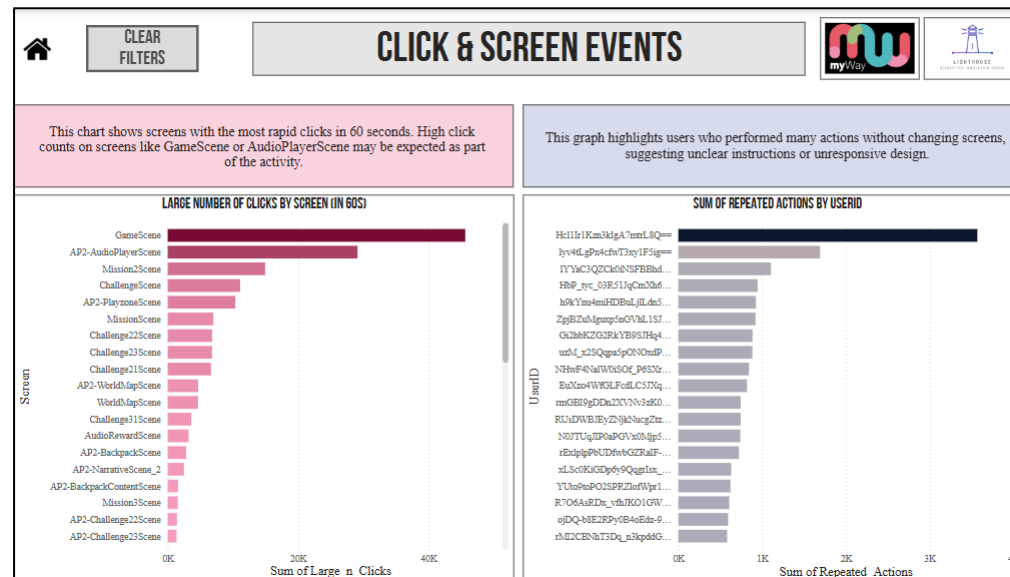
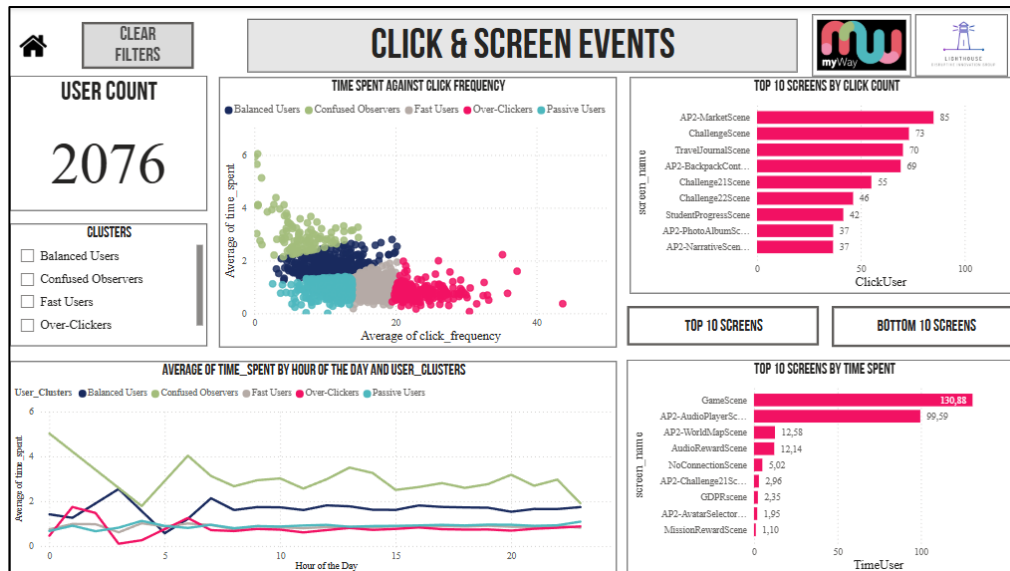
Dataset	Source	Rows	Content
<i>metrics.csv</i>	Lighthouse DIG & Python	2,5M	Click, screen, AI Agent events
<i>activityreports.csv</i>	Lighthouse DIG & Python	250K	Evaluation scores & durations
<i>imgs.txt</i>	Lighthouse DIG	50	Links with screen images
<i>screen_clicks.csv</i>	Lighthouse DIG	600K	Click frequency & screen duration
<i>moods.csv</i>	Metrics.csv & Python	1M	User state
<i>kibot_events.csv</i>	Metrics.csv & Python	200K	AI Agent events
<i>activities_with_moods.csv</i>	Merged Metrics.csv & Activityreports.csv	250K	Scores & moods
<i>kibot_with_grades.csv</i>	Metrics.csv & Kibot_events.csv & Python	1.5K	AI Agent events & scores
<i>screen_summary.csv</i>	Metrics.csv & Python	50	Usability analysis per screen
<i>repeated_uses_summary.csv</i>	Metrics.csv & Python	1.5K	Repetition metrics per user

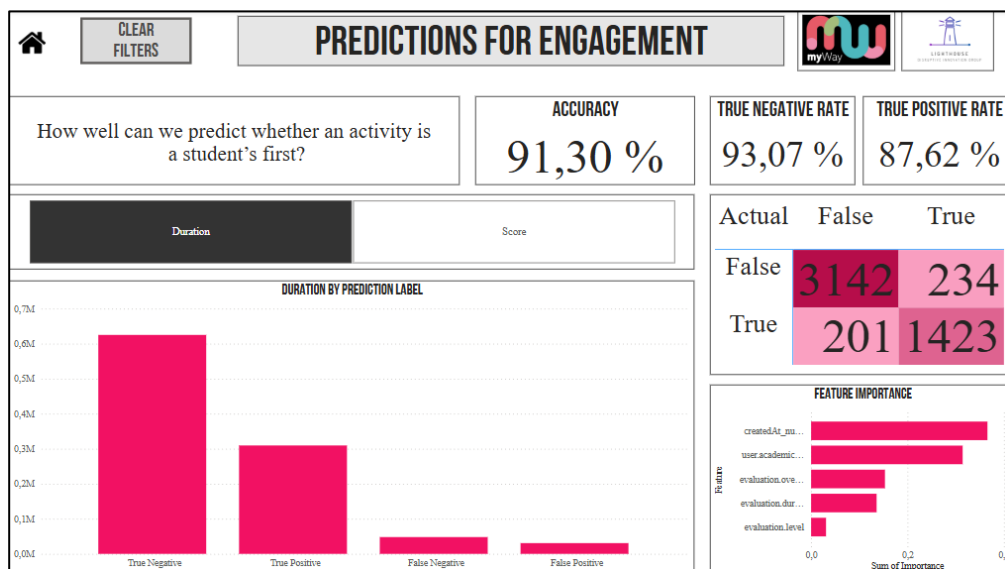
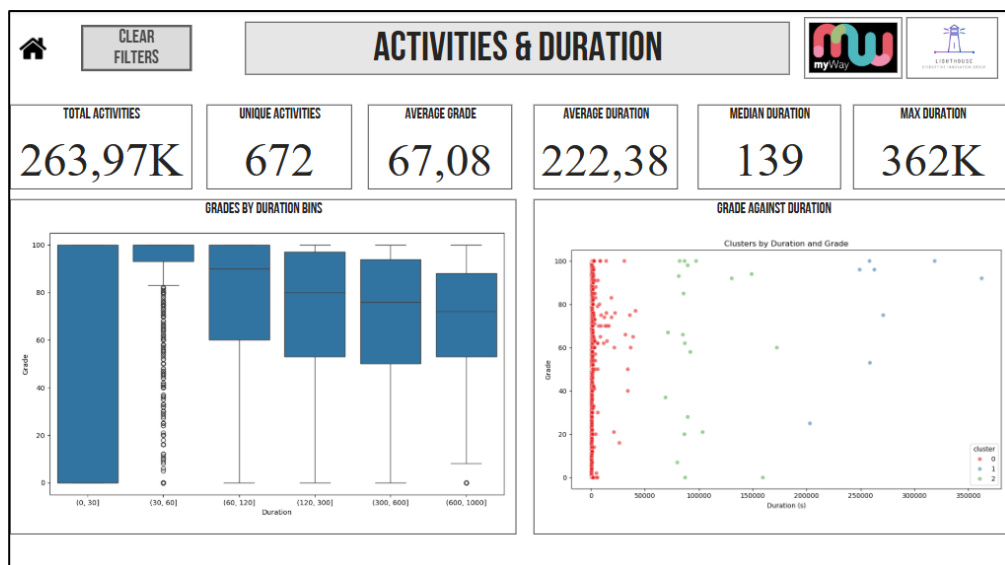
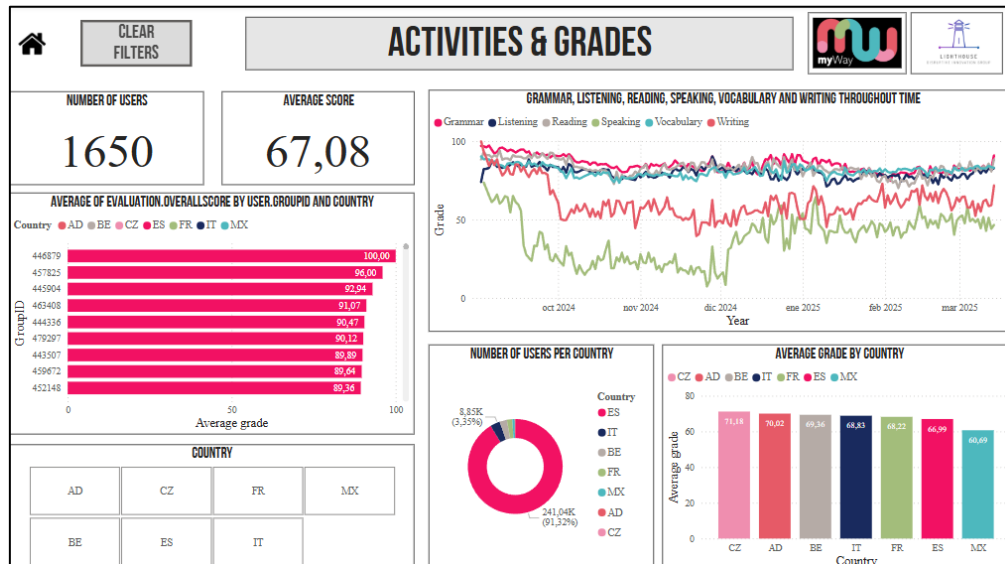
Table 8: Table of Models Comparison

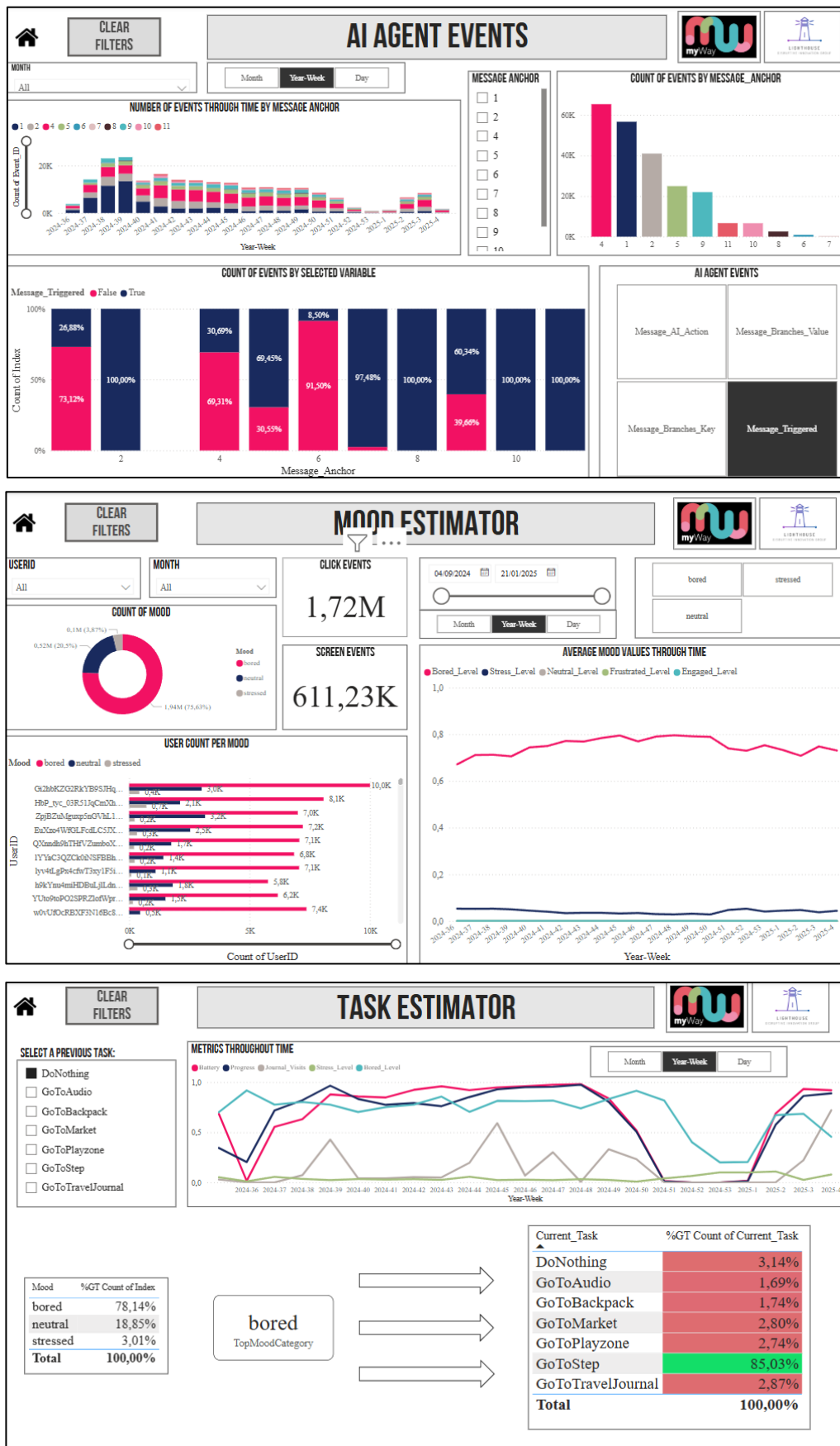
FEATURE SET USED	MODEL 1 – EXTENDED FEATURES	MODEL 2 – CORE FEATURES
Predictors	evaluation.overallScore	
	evaluation.duration	
	evaluation.level	evaluation.overallScore
	user.academicYear	evaluation.duration
	createdAt	evaluation.level
	Engaged_Metric	user.academicYear
	Audio_Progress	createdAt
	Market_Visits	
	Battery	
Model Type	Random Forest (n=20, depth=3)	Random Forest (n=20, depth=3)
Sample Size	5,000 observations	5,000 observations
Accuracy	84%	91,3%

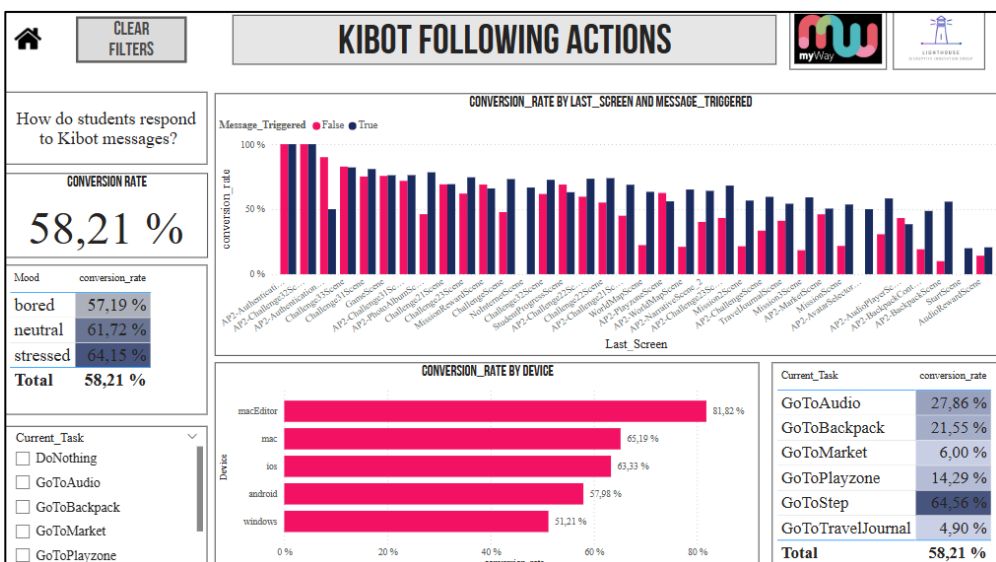
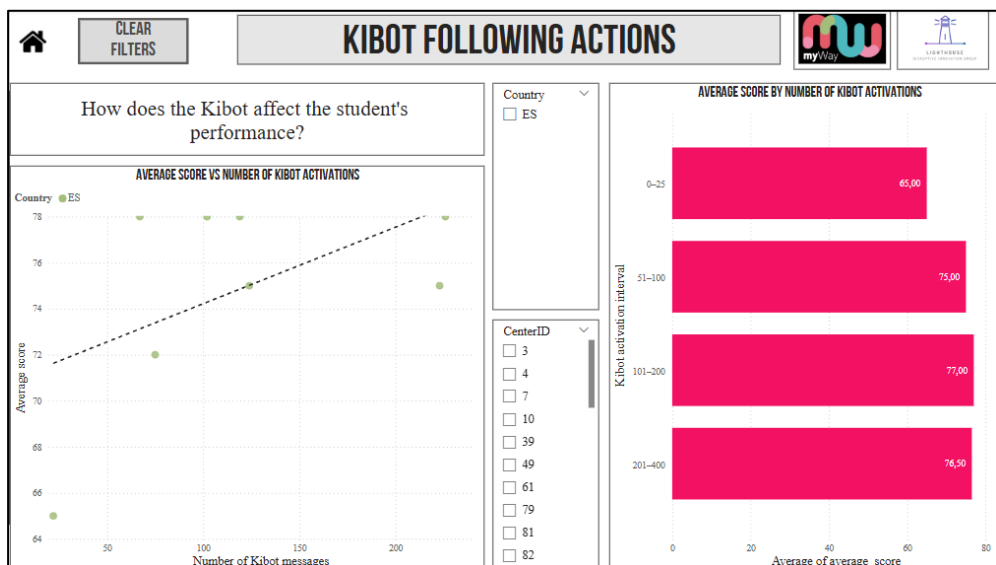
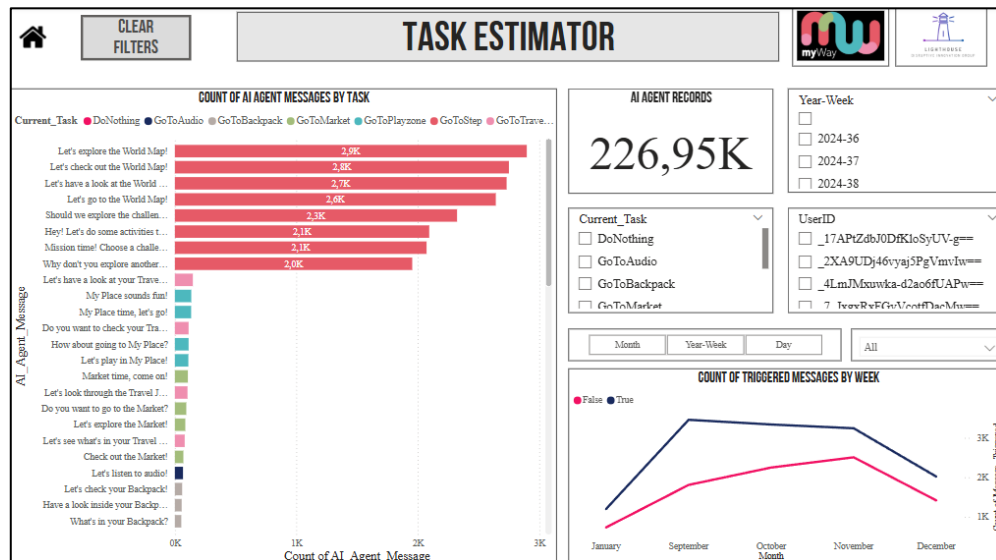
9.3 Appendix D: Power BI pages overview











9.4 Appendix E: GitHub repository

This project was developed using real interaction data provided by Kids&Us. Due to the confidential nature of this data, the real dataset cannot be publicly shared. However, a synthetic dataset was created at the beginning of the project for academic purposes. This dataset replicates the structure, types of variables, and overall patterns of the original data and is sufficient to illustrate the methodology and processes used throughout the project.

All code and deliverables, including the final Power BI dashboard, are stored in a GitHub repository. This repository is set to private to protect sensitive content. Access has been granted to the project supervisors via GitHub. In the case of access issues, a screenshot of the repository's structure is included below for reference, and a Google Drive folder with the same contents is provided as a backup.

GitHub Repository (Private):

<https://github.com/mariachristog/Maria-TFG>

(Access granted to reviewers. Please contact the author if access is not working.)

Backup Google Drive Folder:

https://drive.google.com/drive/folders/1FeykU82Ugidxy5XOKOyyclNbI7aefNMI?usp=drive_link

(Includes the same files as the GitHub repository for reviewer convenience.)

Repository Structure Screenshot:

